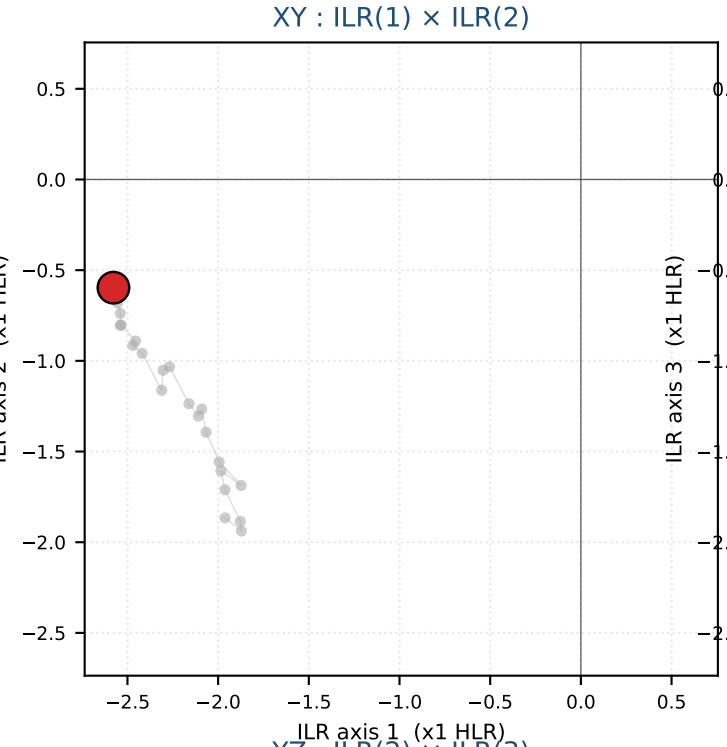
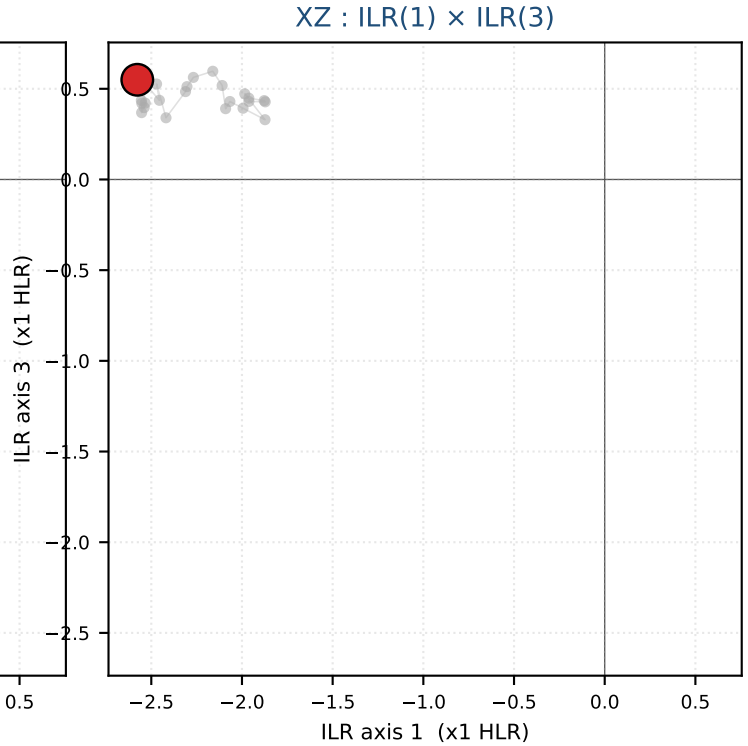
STAGE 1 v4 | t=0 | 2001 | D=9 T=25 | level 1 first-principles

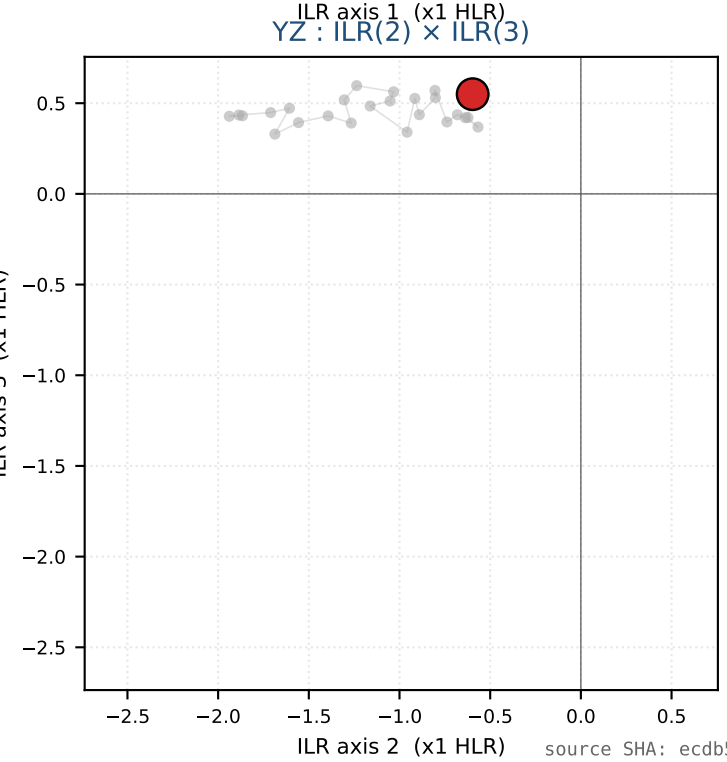
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

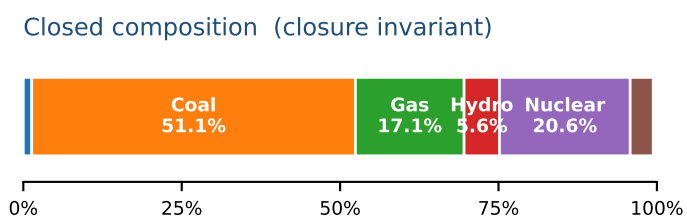




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0133	+3.365
Coal	0.5109	+7.009
Gas	0.1715	+5.918
Hydro	0.0558	+4.796
Nuclear	0.2063	+6.102
Other Fossil	0.0364	+4.367
Other Renewabl	0.0040	+2.158
Solar	0.0000	-35.081
Wind	0.0018	+1.366





closure: Σ = 1.000000 OK

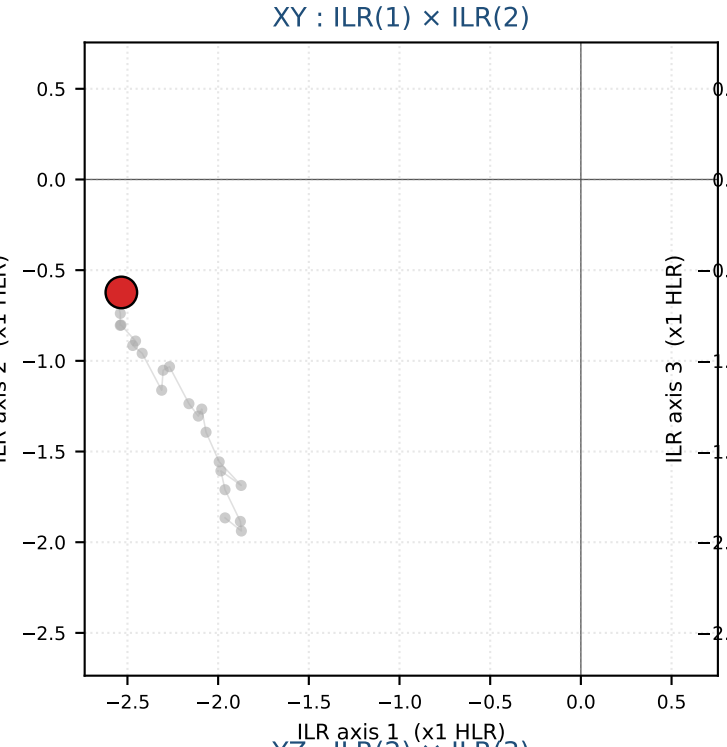
display factor: x1 (raw HLR x 1)

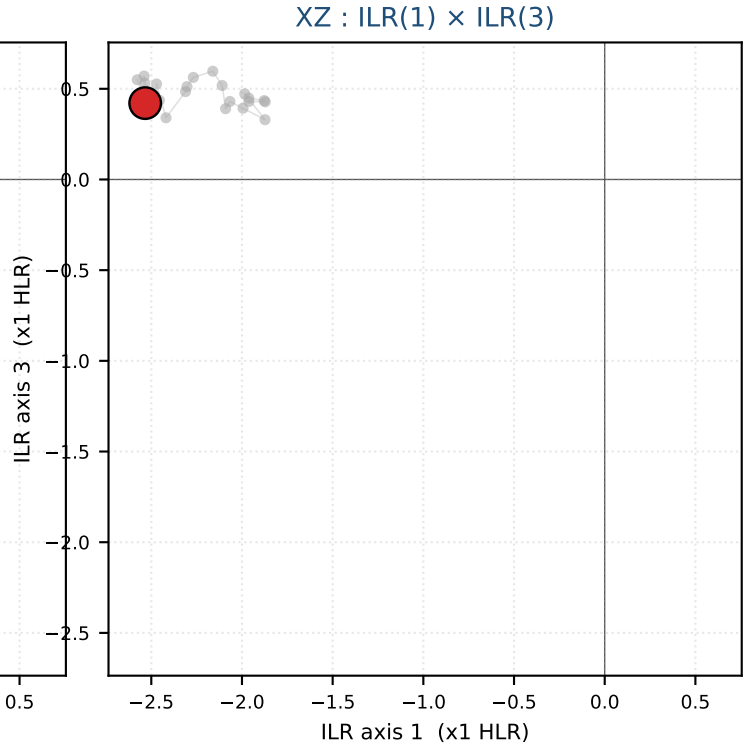
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=1 | 2002 | D=9 T=25 | level 1 first-principles

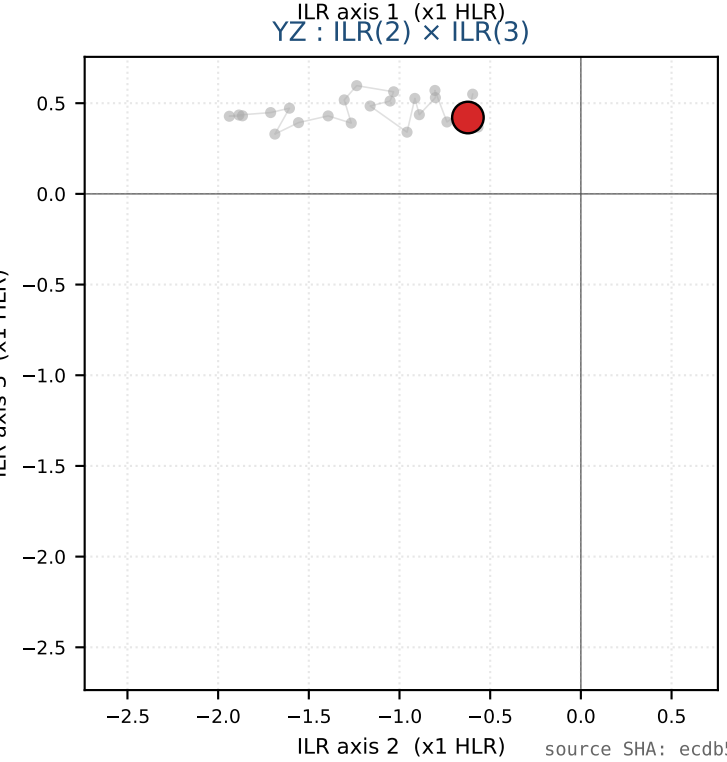
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



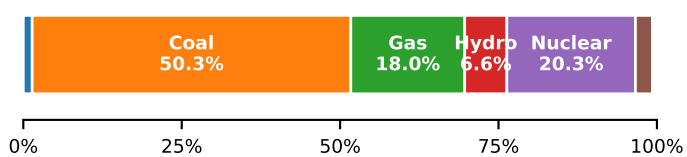


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0140	+3.371
Coal	0.5029	+6.954
Gas	0.1798	+5.926
Hydro	0.0665	+4.931
Nuclear	0.2029	+6.047
Other Fossil	0.0270	+4.029
Other Renewabl	0.0042	+2.168
Solar	0.0000	-35.151
Wind	0.0027	+1.725



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

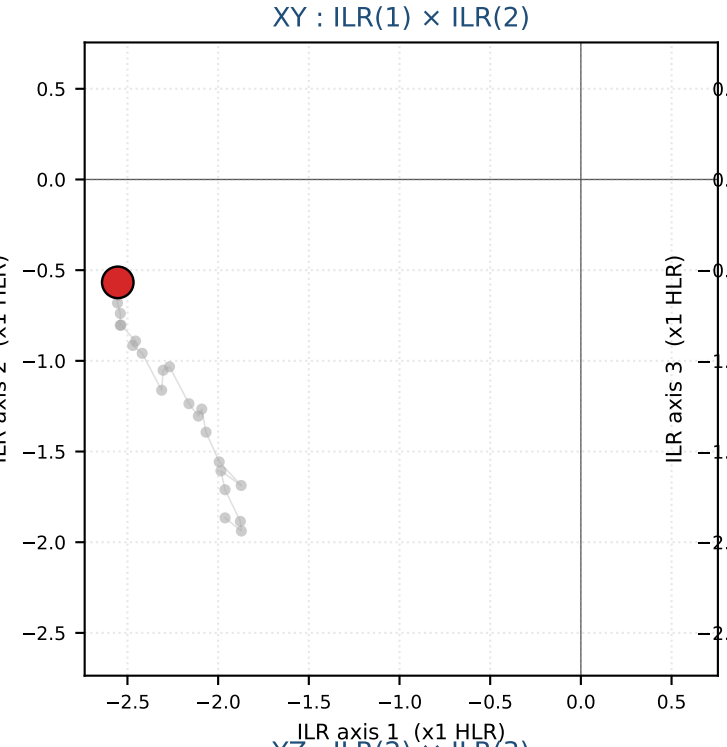
display factor: x1 (raw HLR x 1)

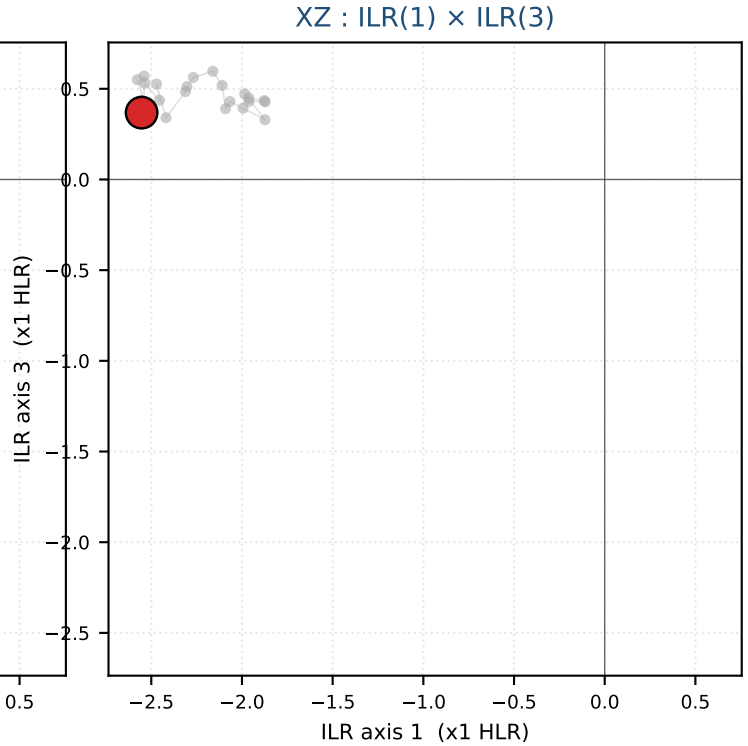
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=2 | 2003 | D=9 T=25 | level 1 first-principles

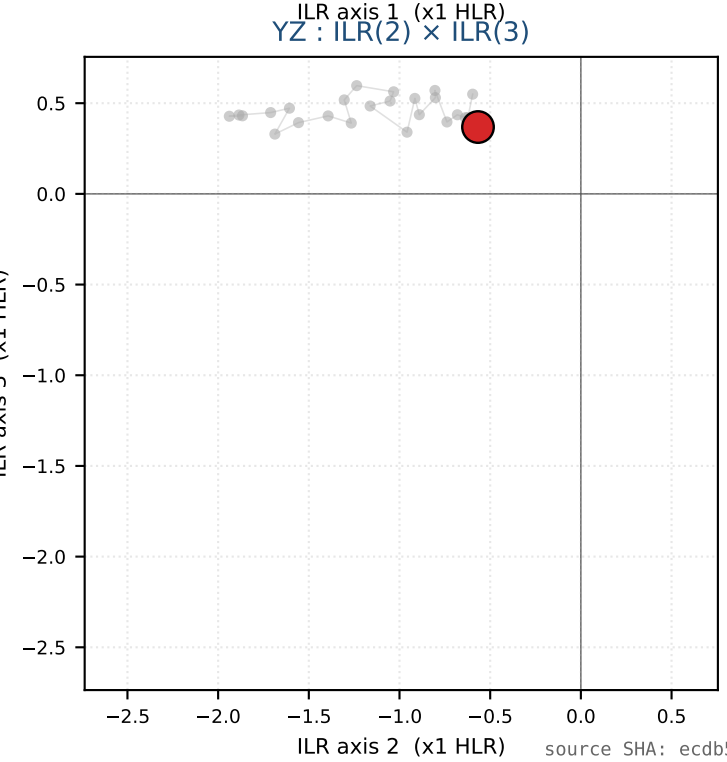
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

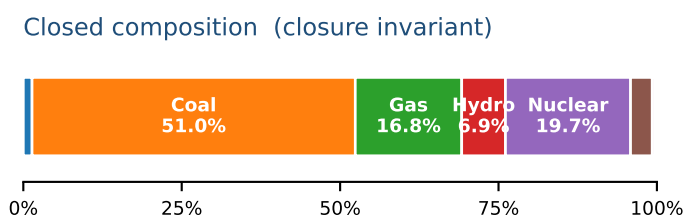




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0138	-0.441
Coal	0.5101	+3.170
Gas	0.1680	+2.059
Hydro	0.0691	+1.171
Nuclear	0.1974	+2.221
Other Fossil	0.0342	+0.468
Other Renewabl	0.0044	-1.593
Solar	0.0001	-5.053
Wind	0.0029	-2.002





closure: Σ = 1.000000 OK

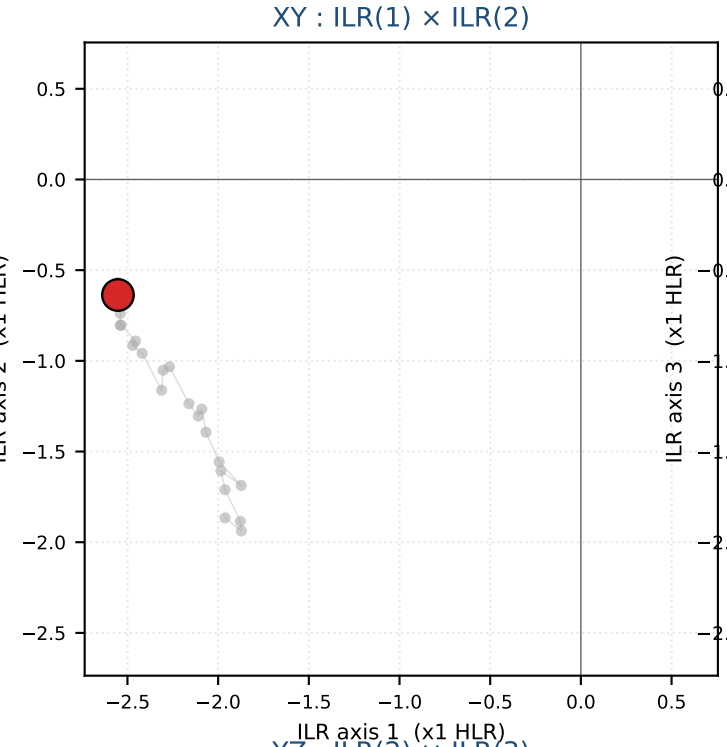
display factor: x1 (raw HLR x 1)

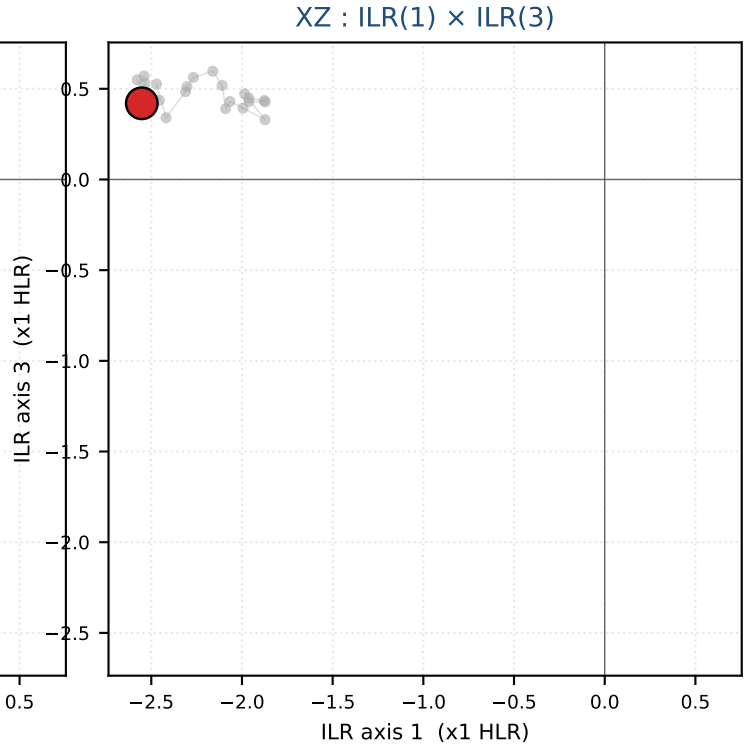
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=3 | 2004 | D=9 T=25 | level 1 first-principles

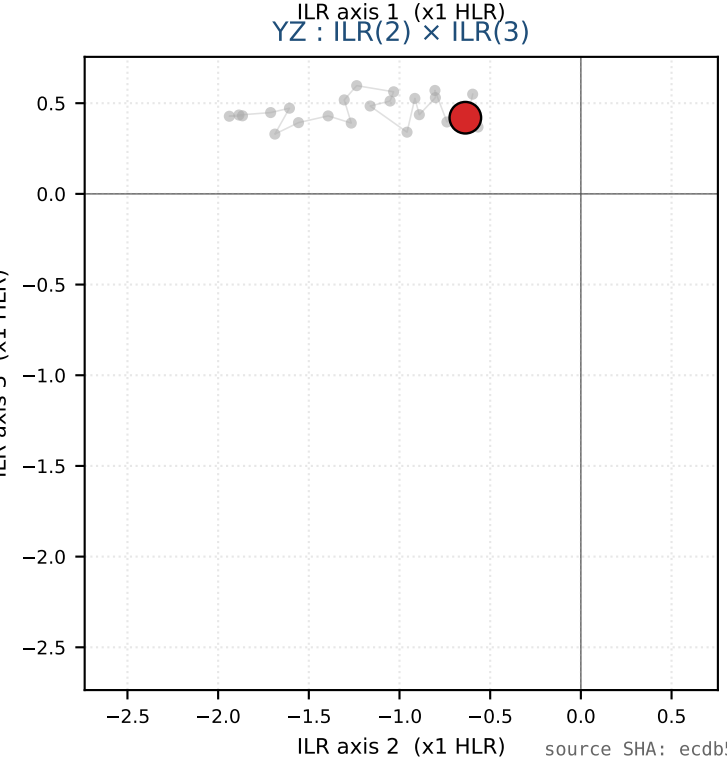
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



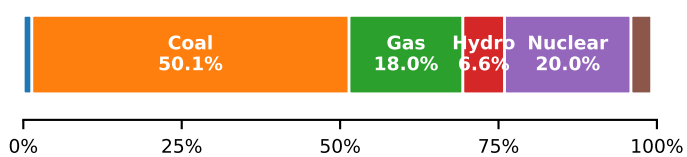


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0135	-0.481
Coal	0.5006	+3.129
Gas	0.1797	+2.104
Hydro	0.0658	+1.099
Nuclear	0.1995	+2.209
Other Fossil	0.0329	+0.405
Other Renewabl	0.0043	-1.631
Solar	0.0001	-5.023
Wind	0.0036	-1.813



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

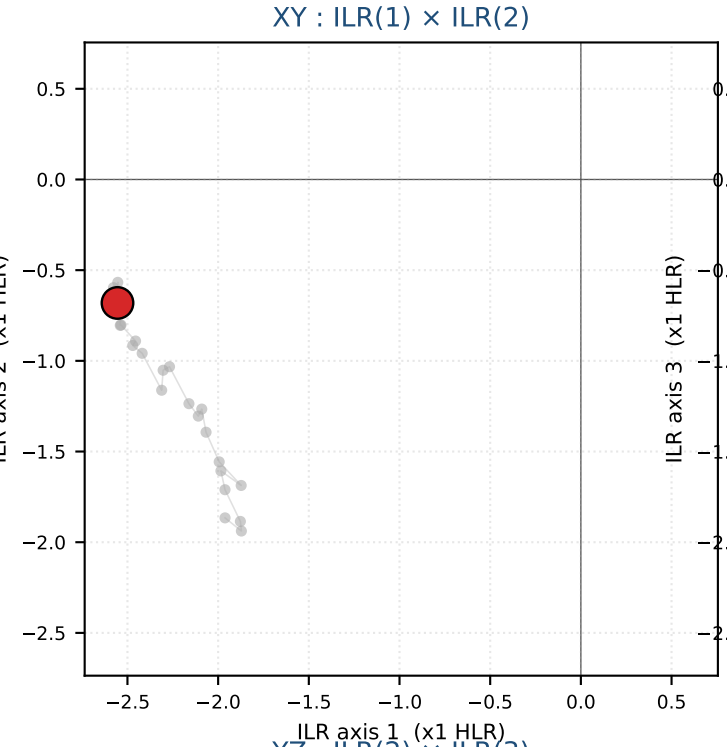
display factor: x1 (raw HLR x 1)

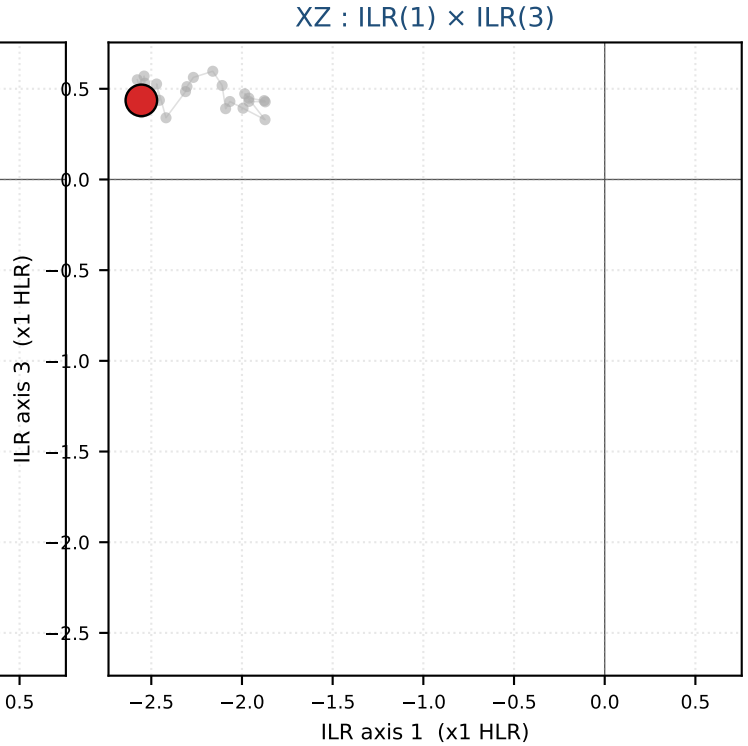
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=4 | 2005 | D=9 T=25 | level 1 first-principles

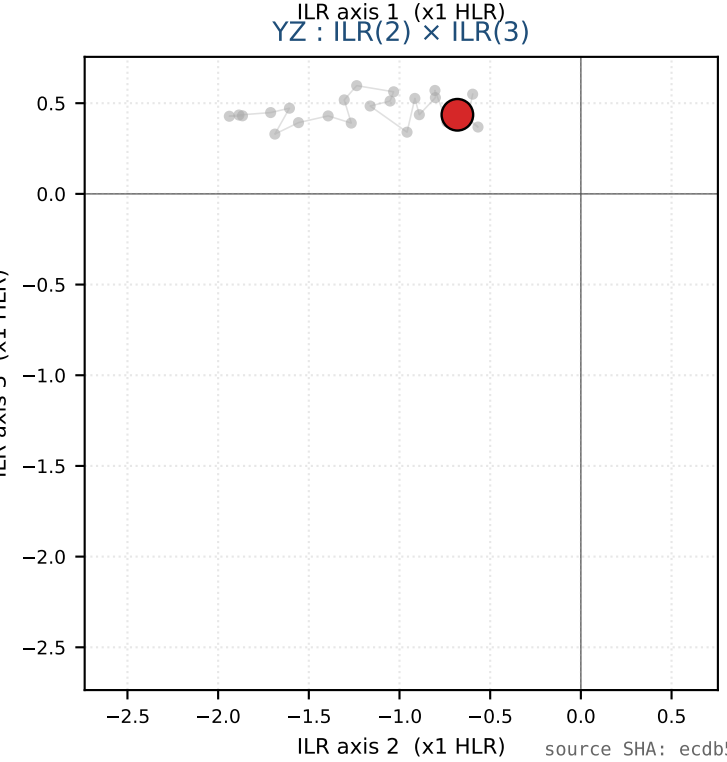
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

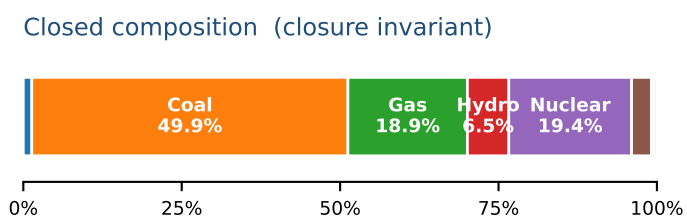




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0135	-0.495
Coal	0.4989	+3.118
Gas	0.1886	+2.145
Hydro	0.0654	+1.086
Nuclear	0.1938	+2.172
Other Fossil	0.0313	+0.348
Other Renewabl	0.0041	-1.678
Solar	0.0001	-5.087
Wind	0.0044	-1.609





closure: Σ = 1.000000 OK

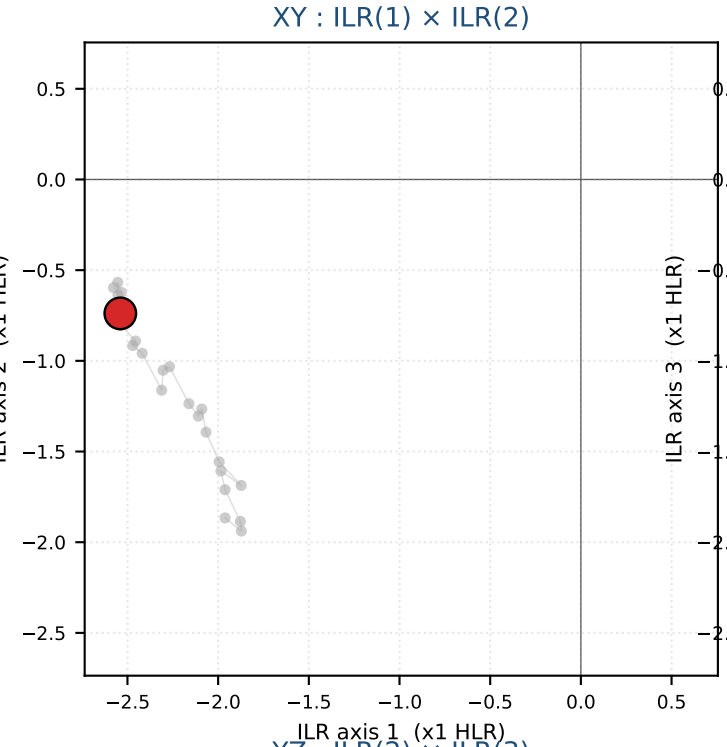
display factor: x1 (raw HLR x 1)

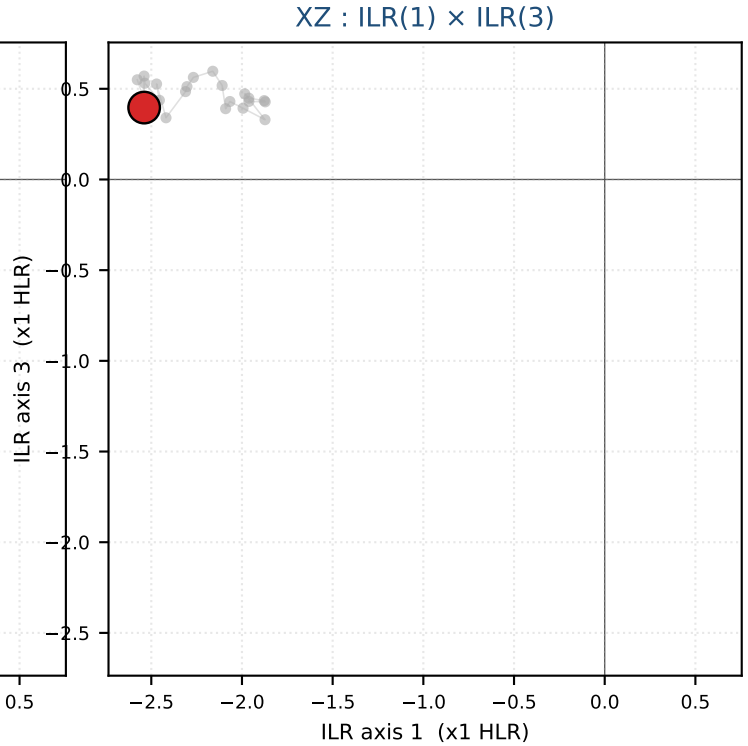
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=5 | 2006 | D=9 T=25 | level 1 first-principles

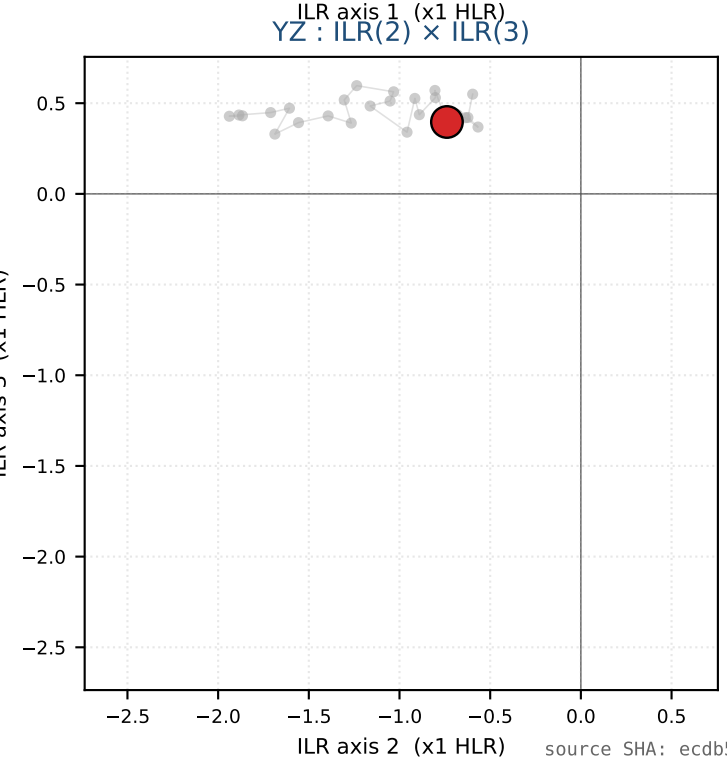
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



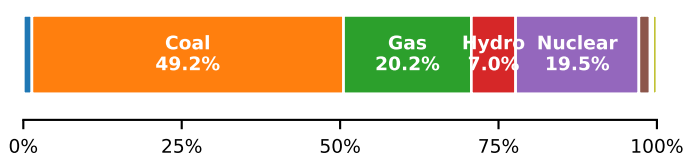


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0136	-0.469
Coal	0.4919	+3.123
Gas	0.2017	+2.231
Hydro	0.0699	+1.171
Nuclear	0.1945	+2.195
Other Fossil	0.0177	-0.202
Other Renewabl	0.0041	-1.670
Solar	0.0001	-5.187
Wind	0.0066	-1.193



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

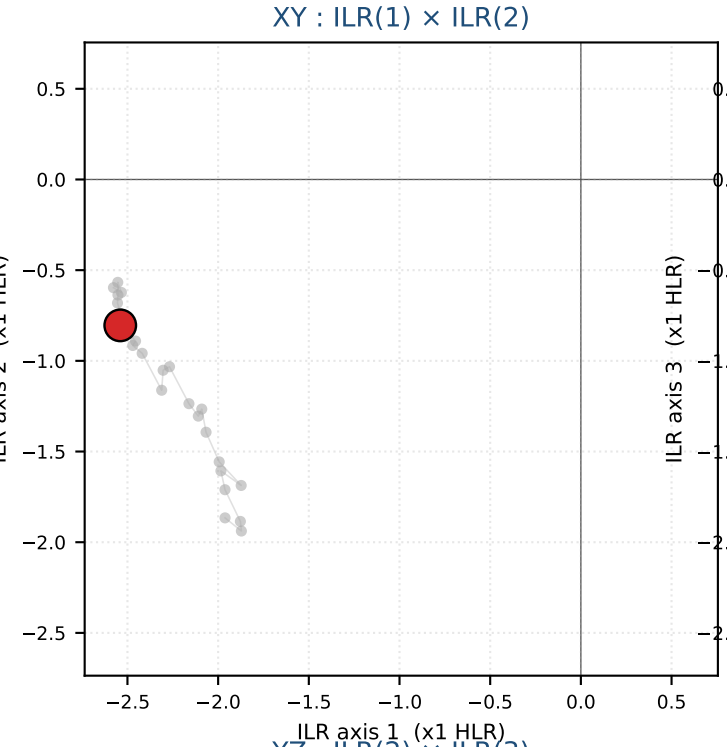
display factor: x1 (raw HLR x 1)

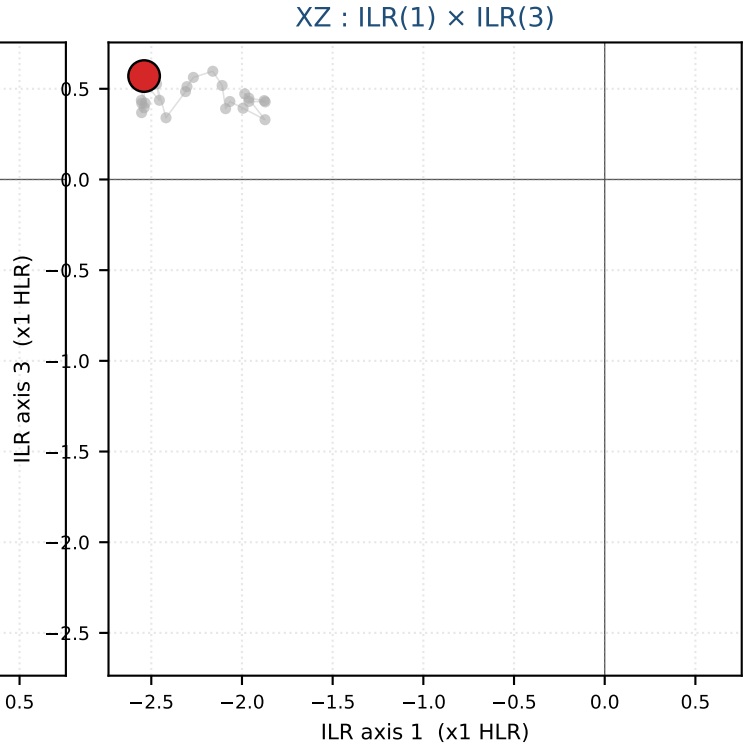
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=6 | 2007 | D=9 T=25 | level 1 first-principles

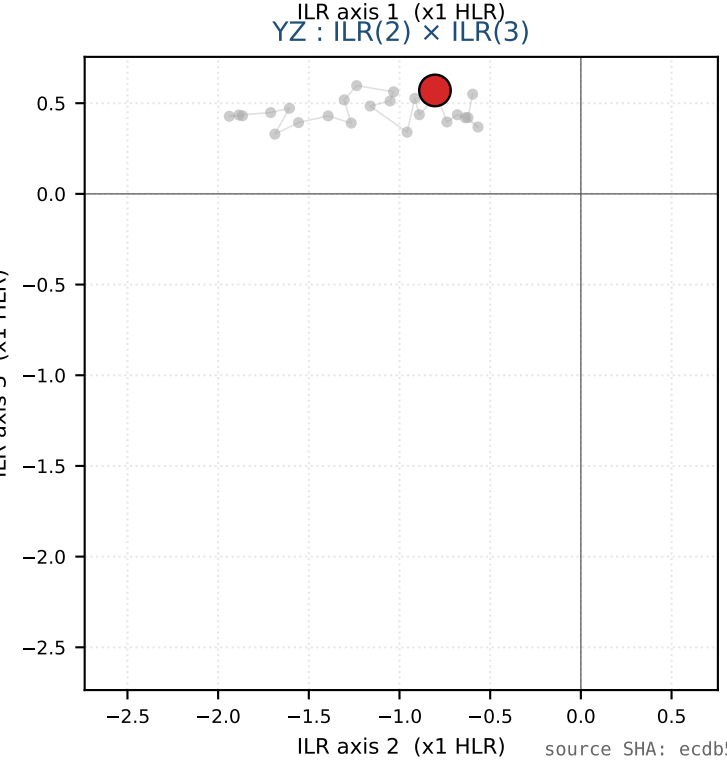
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

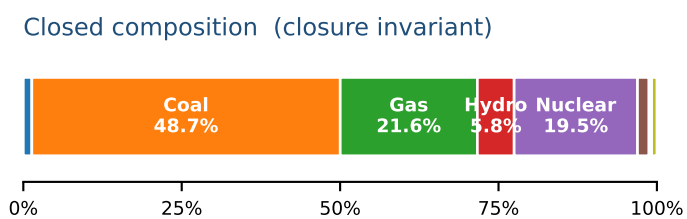




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0134	-0.514
Coal	0.4868	+3.078
Gas	0.2164	+2.267
Hydro	0.0581	+0.952
Nuclear	0.1947	+2.161
Other Fossil	0.0182	-0.211
Other Renewabl	0.0040	-1.715
Solar	0.0001	-5.026
Wind	0.0083	-0.992





closure: Σ = 1.000000 OK

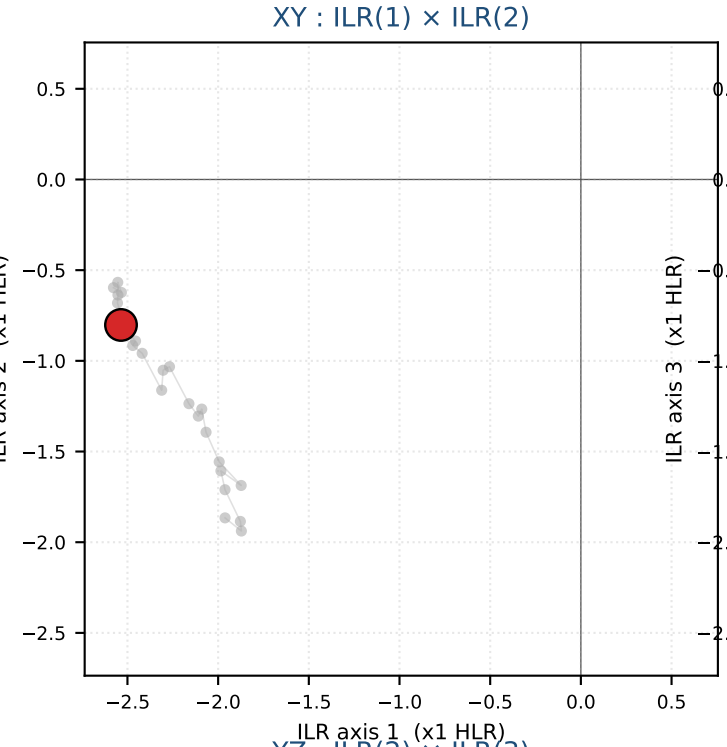
display factor: x1 (raw HLR x 1)

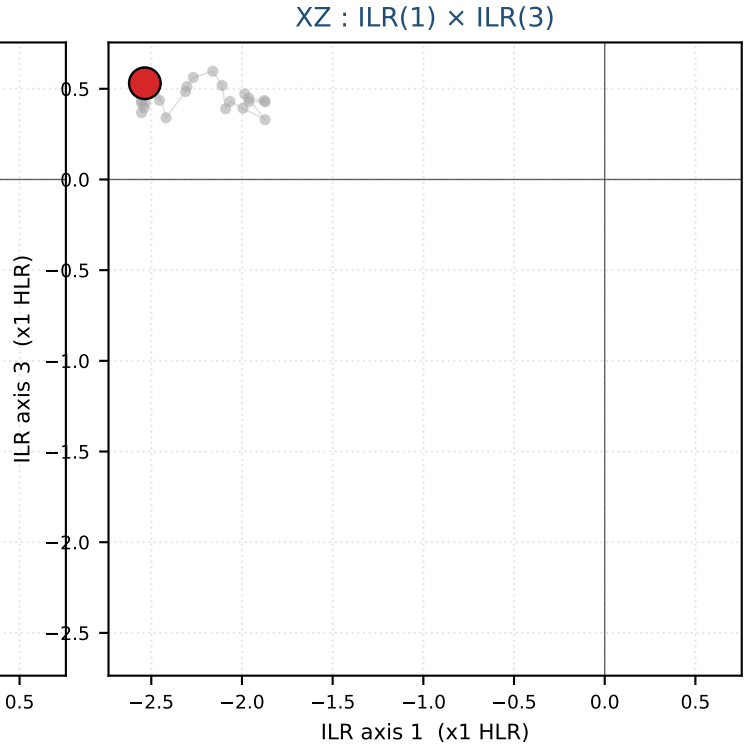
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=7 | 2008 | D=9 T=25 | level 1 first-principles

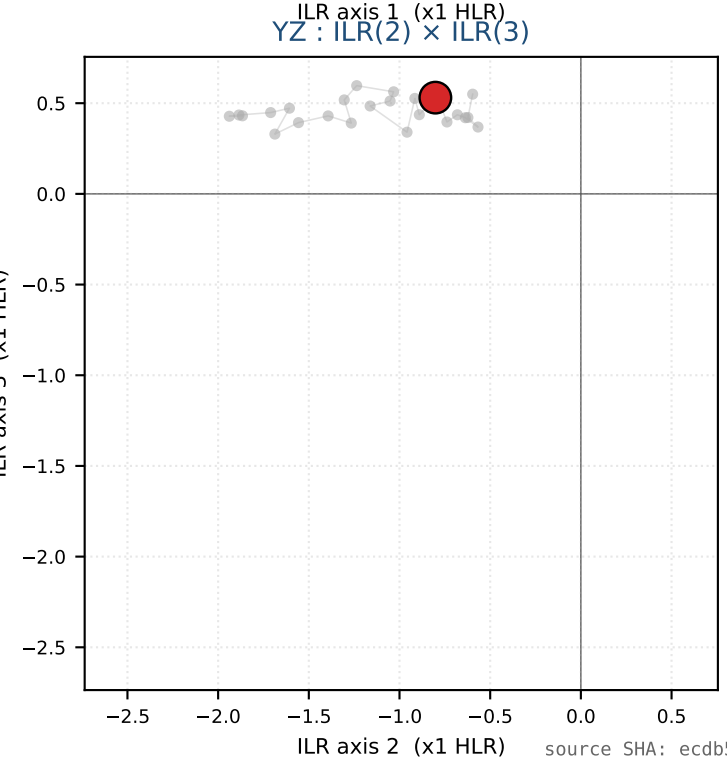
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

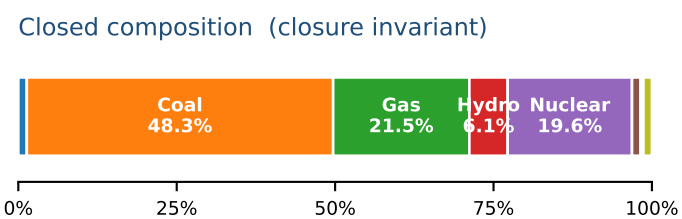




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0134	-0.585
Coal	0.4835	+3.001
Gas	0.2150	+2.191
Hydro	0.0605	+0.923
Nuclear	0.1963	+2.100
Other Fossil	0.0135	-0.578
Other Renewabl	0.0042	-1.741
Solar	0.0002	-4.732
Wind	0.0135	-0.579





closure: Σ = 1.000000 OK

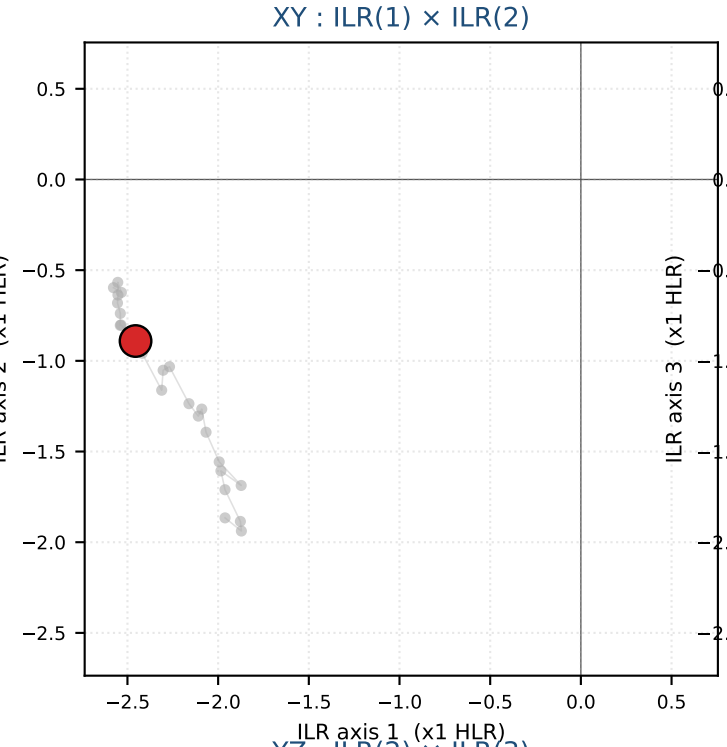
display factor: x1 (raw HLR x 1)

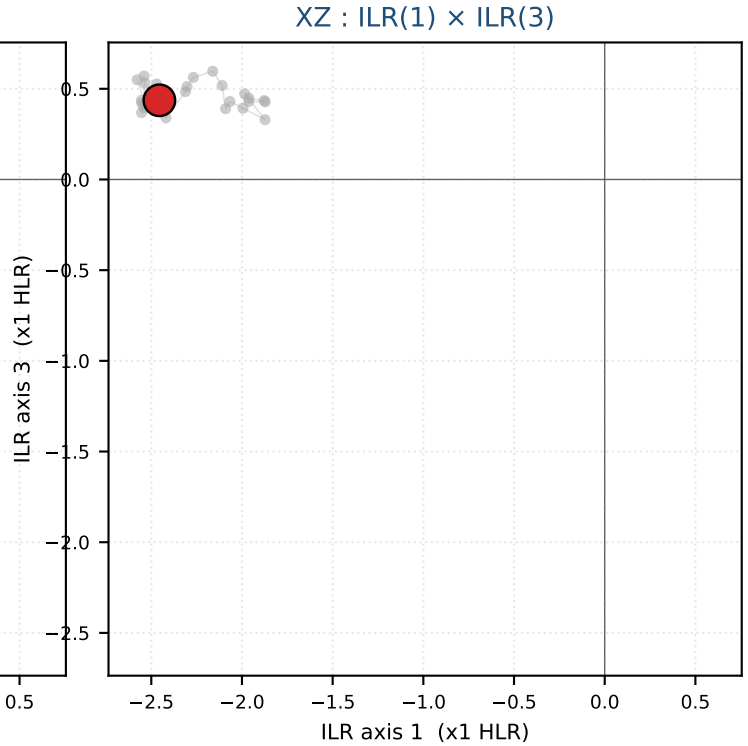
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=8 | 2009 | D=9 T=25 | level 1 first-principles

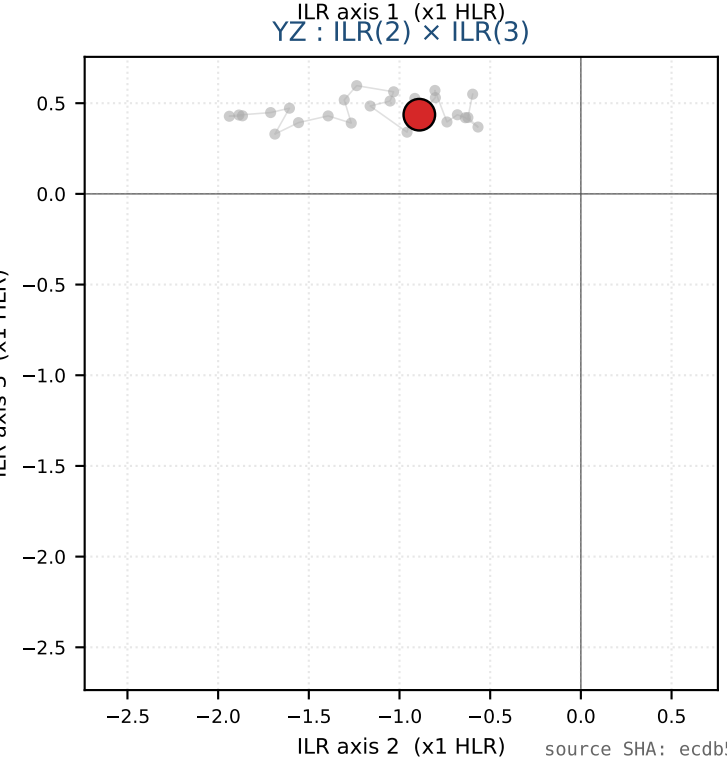
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

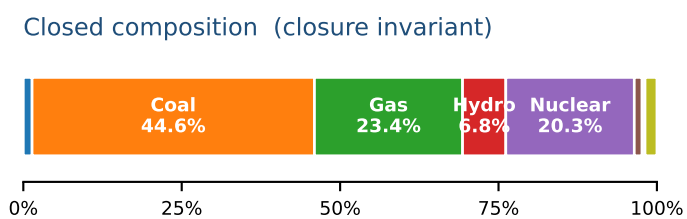




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0138	-0.613
Coal	0.4457	+2.860
Gas	0.2338	+2.214
Hydro	0.0682	+0.983
Nuclear	0.2028	+2.072
Other Fossil	0.0123	-0.729
Other Renewabl	0.0044	-1.751
Solar	0.0002	-4.728
Wind	0.0188	-0.309





closure: Σ = 1.000000 OK

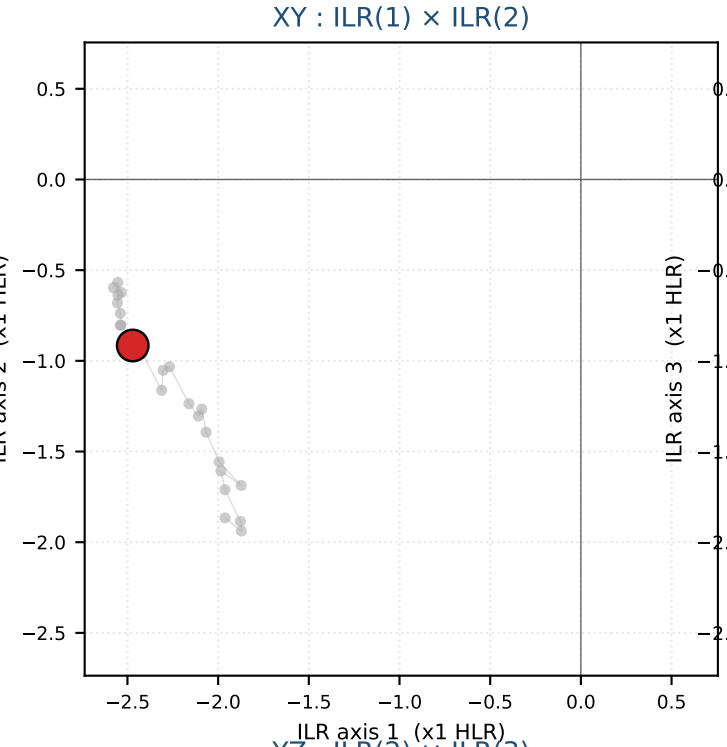
display factor: x1 (raw HLR x 1)

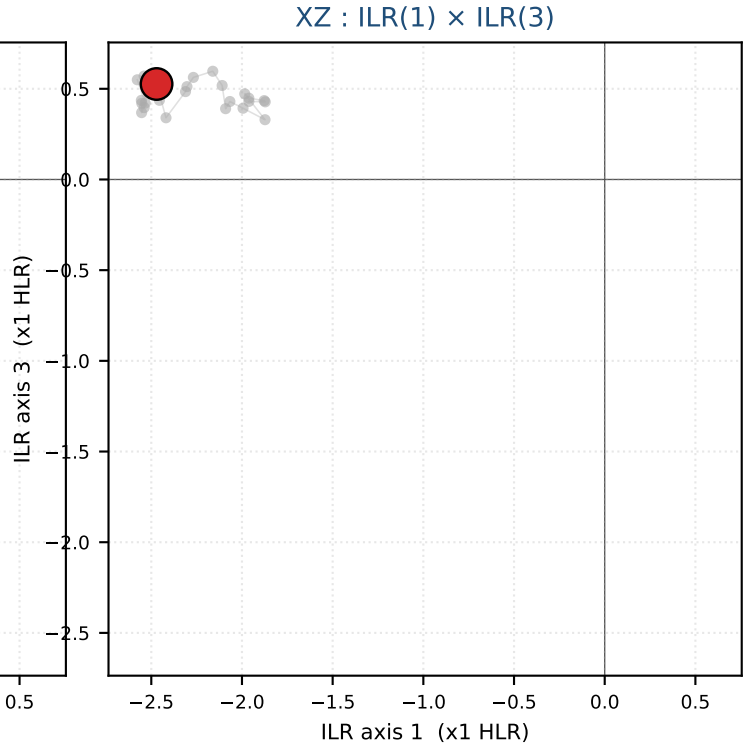
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=9 | 2010 | D=9 T=25 | level 1 first-principles

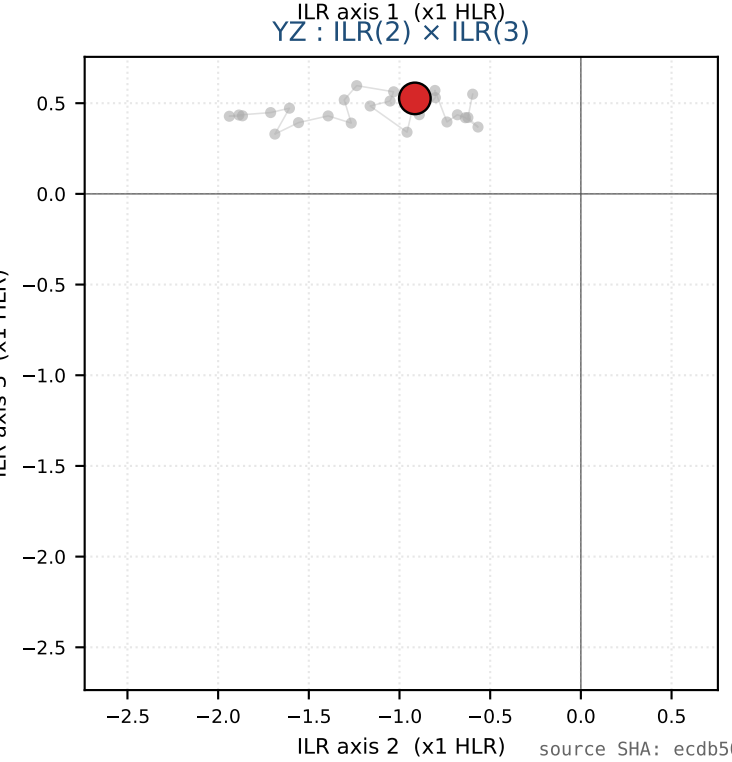
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



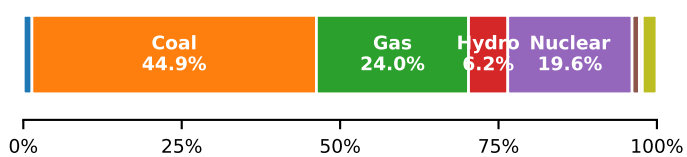


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0136	-0.658
Coal	0.4490	+2.836
Gas	0.2401	+2.210
Hydro	0.0619	+0.855
Nuclear	0.1962	+2.008
Other Fossil	0.0116	-0.824
Other Renewabl	0.0043	-1.804
Solar	0.0003	-4.487
Wind	0.0230	-0.135



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

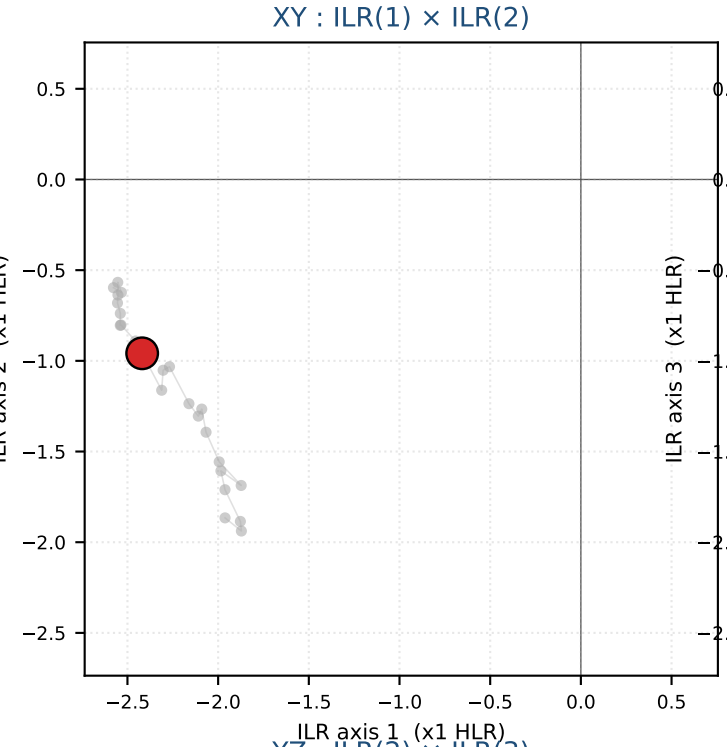
display factor: x1 (raw HLR x 1)

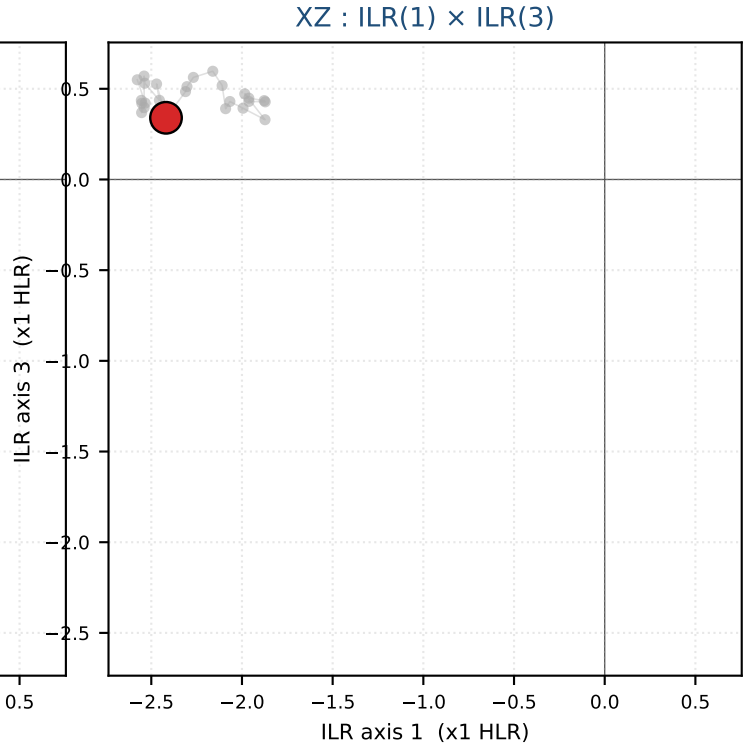
Helmert basis loadings (top 3 carriers per axis)

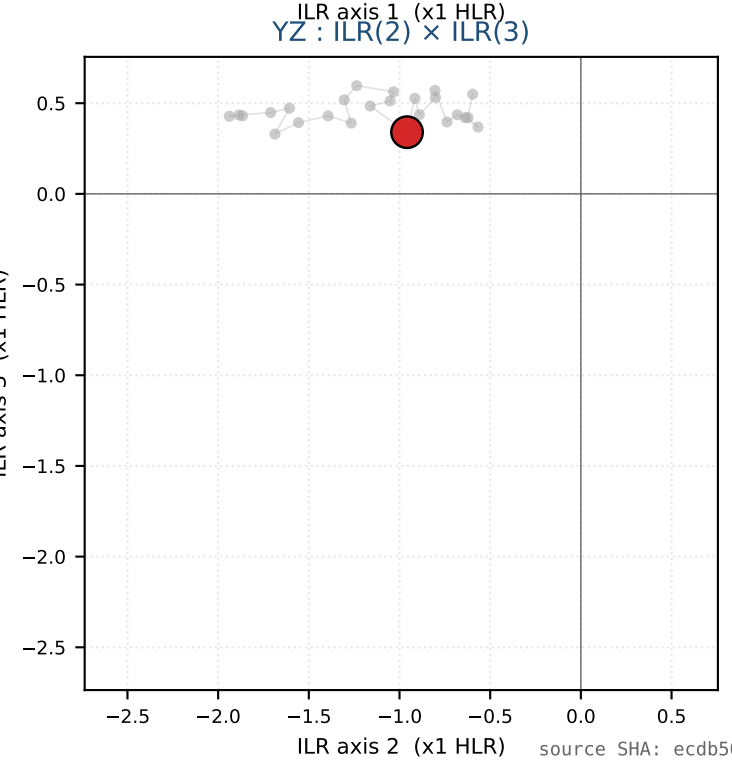
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=10 | 2011 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



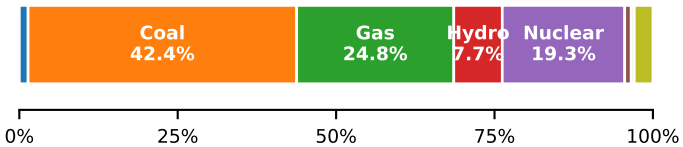




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0139	-0.725
Coal	0.4239	+2.696
Gas	0.2479	+2.160
Hydro	0.0765	+0.984
Nuclear	0.1933	+1.911
Other Fossil	0.0102	-1.029
Other Renewabl	0.0044	-1.862
Solar	0.0004	-4.163
Wind	0.0294	+0.027

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

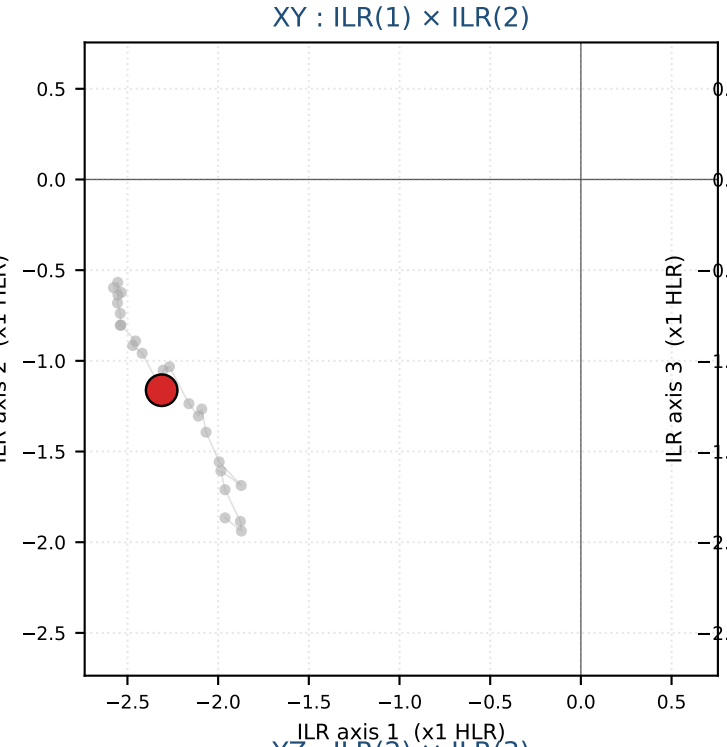
display factor: x1 (raw HLR x 1)

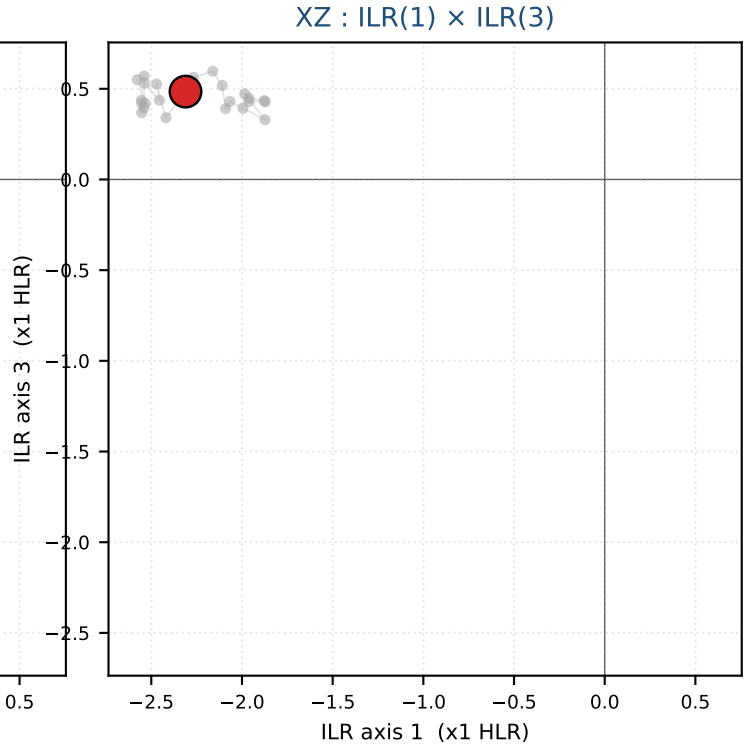
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=11 | 2012 | D=9 T=25 | level 1 first-principles

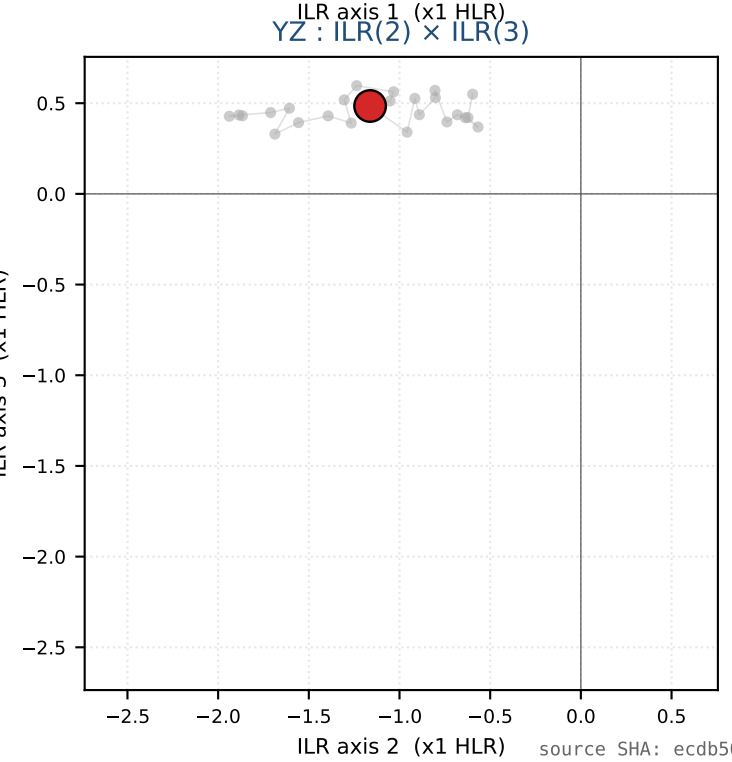
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

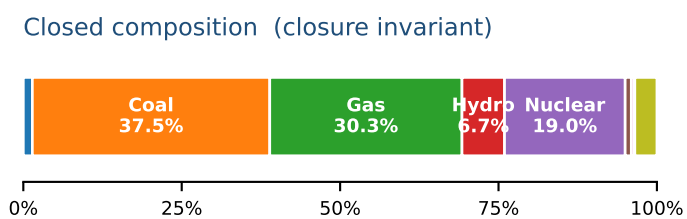




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0143	-0.804
Coal	0.3747	+2.465
Gas	0.3034	+2.253
Hydro	0.0671	+0.745
Nuclear	0.1904	+1.788
Other Fossil	0.0097	-1.193
Other Renewabl	0.0045	-1.953
Solar	0.0011	-3.390
Wind	0.0349	+0.089





closure: Σ = 1.000000 OK

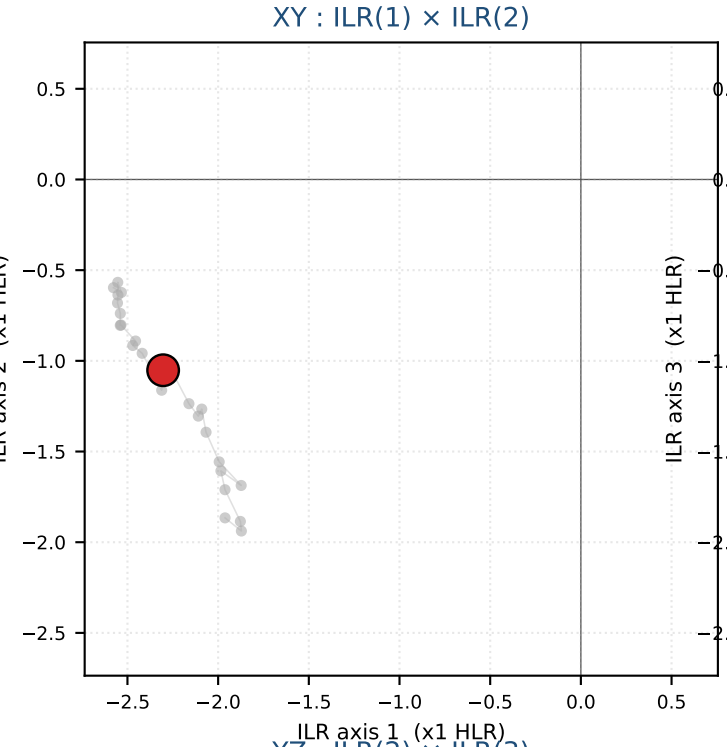
display factor: x1 (raw HLR x 1)

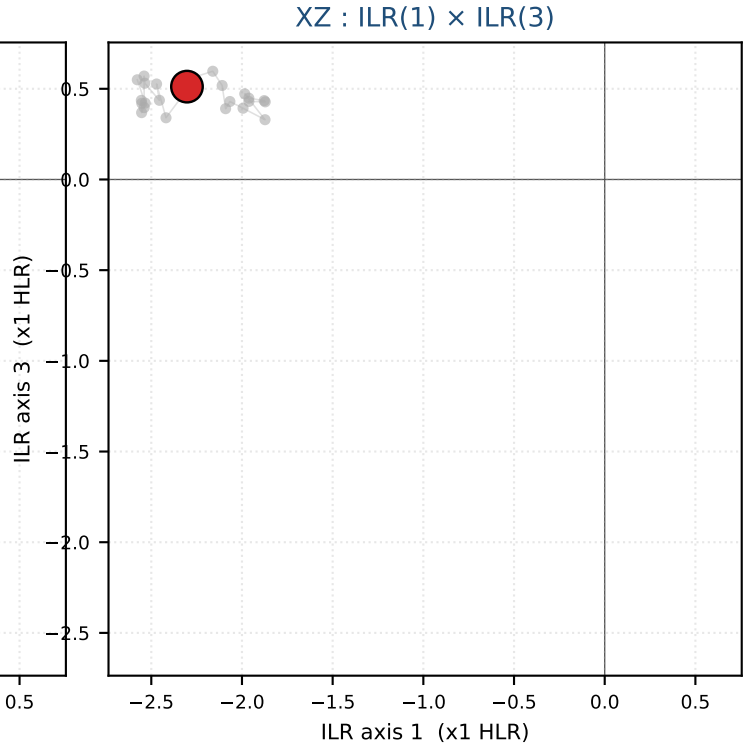
Helmert basis loadings (top 3 carriers per axis)

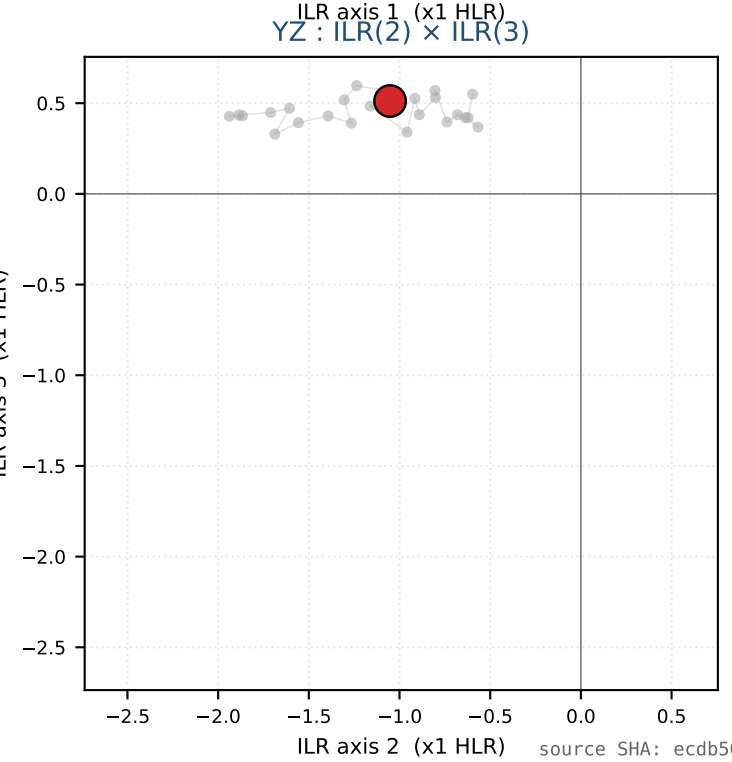
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=12 | 2013 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



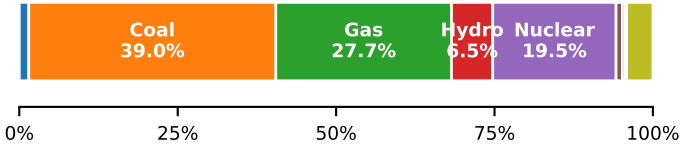




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0150	-0.858
Coal	0.3899	+2.400
Gas	0.2774	+2.059
Hydro	0.0651	+0.609
Nuclear	0.1945	+1.704
Other Fossil	0.0099	-1.271
Other Renewabl	0.0046	-2.035
Solar	0.0022	-2.766
Wind	0.0414	+0.157

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

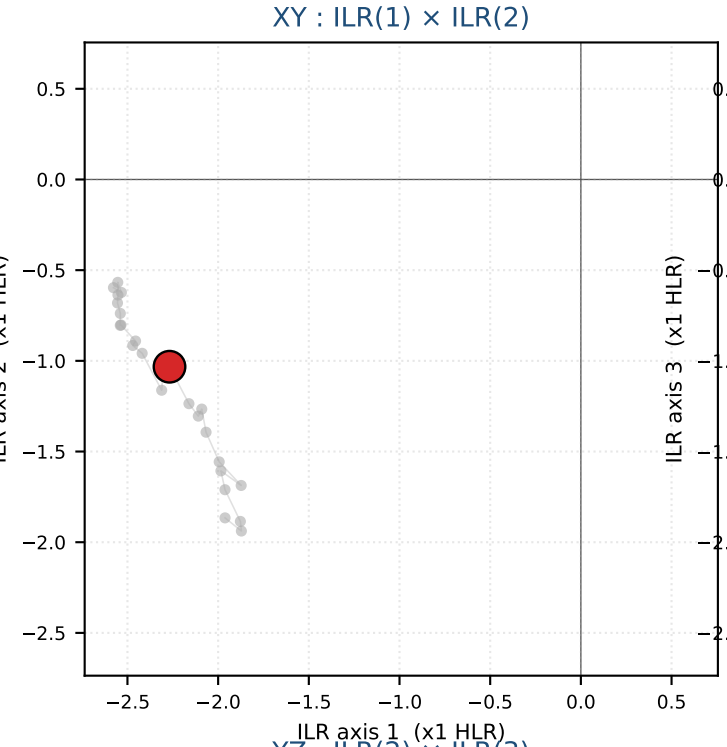
display factor: x1 (raw HLR x 1)

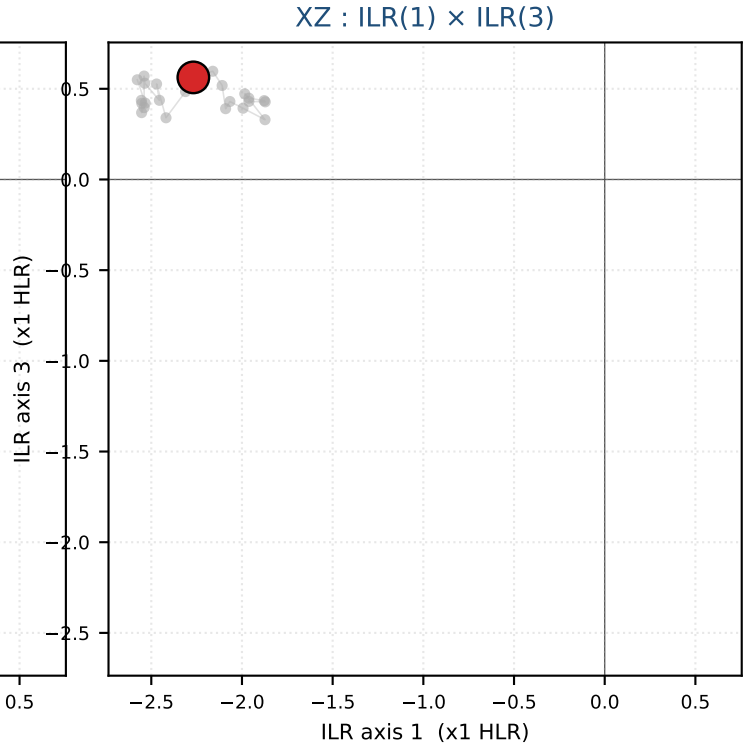
Helmert basis loadings (top 3 carriers per axis)

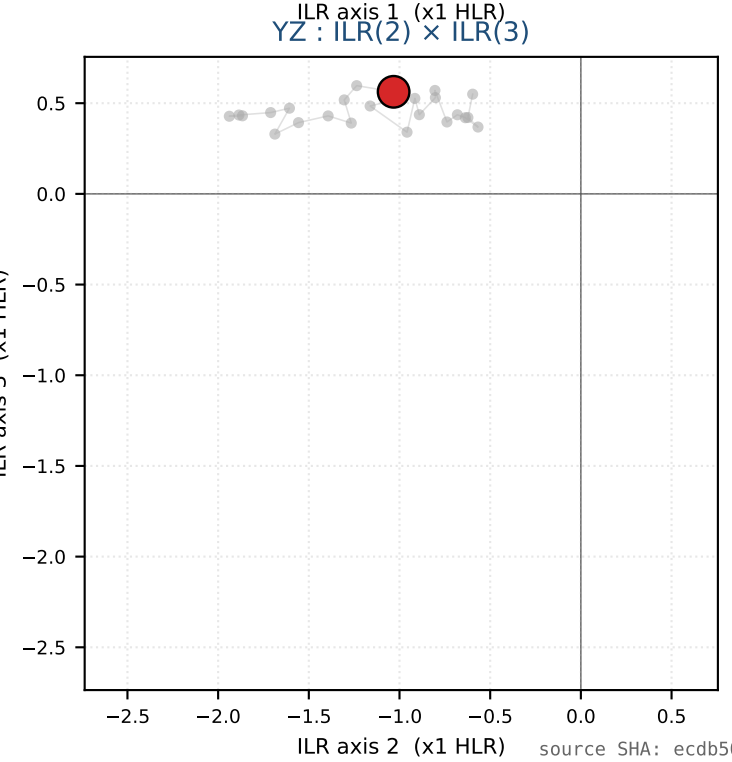
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=13 | 2014 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



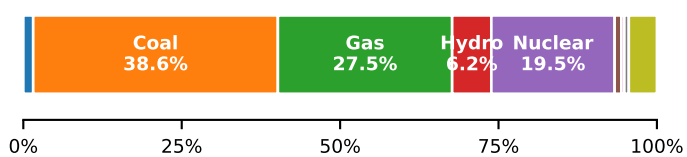




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0156	-0.959
Coal	0.3862	+2.249
Gas	0.2750	+1.909
Hydro	0.0618	+0.417
Nuclear	0.1946	+1.563
Other Fossil	0.0107	-1.340
Other Renewabl	0.0047	-2.171
Solar	0.0071	-1.753
Wind	0.0444	+0.085

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

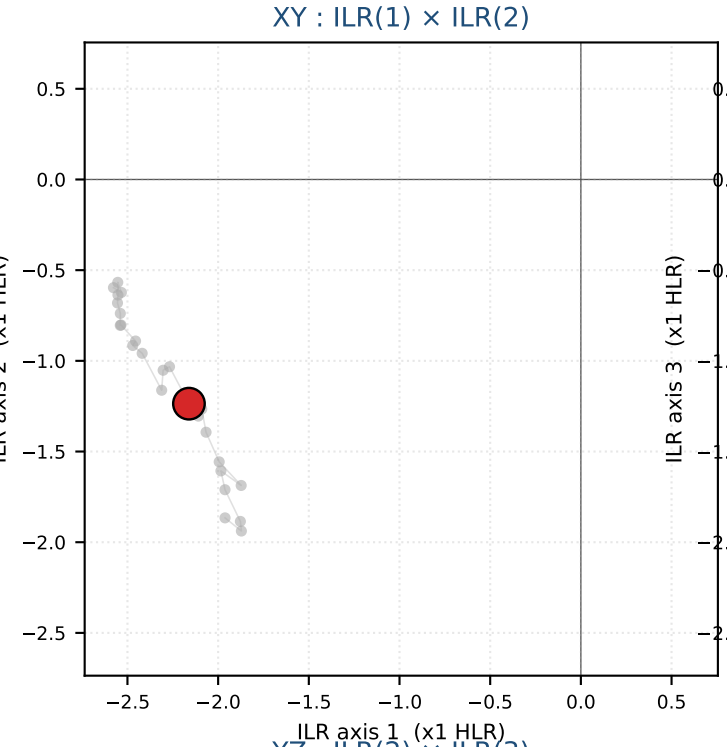
display factor: x1 (raw HLR x 1)

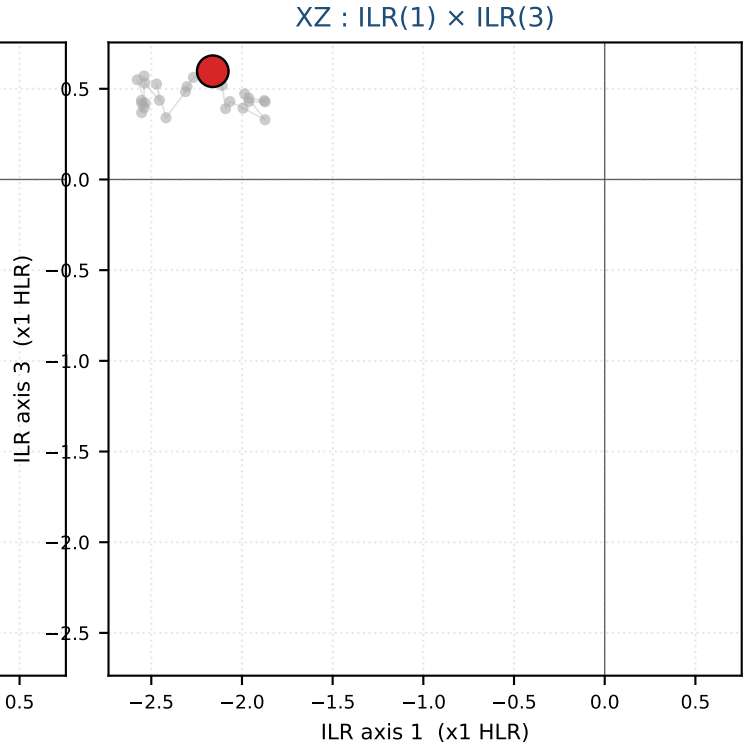
Helmert basis loadings (top 3 carriers per axis)

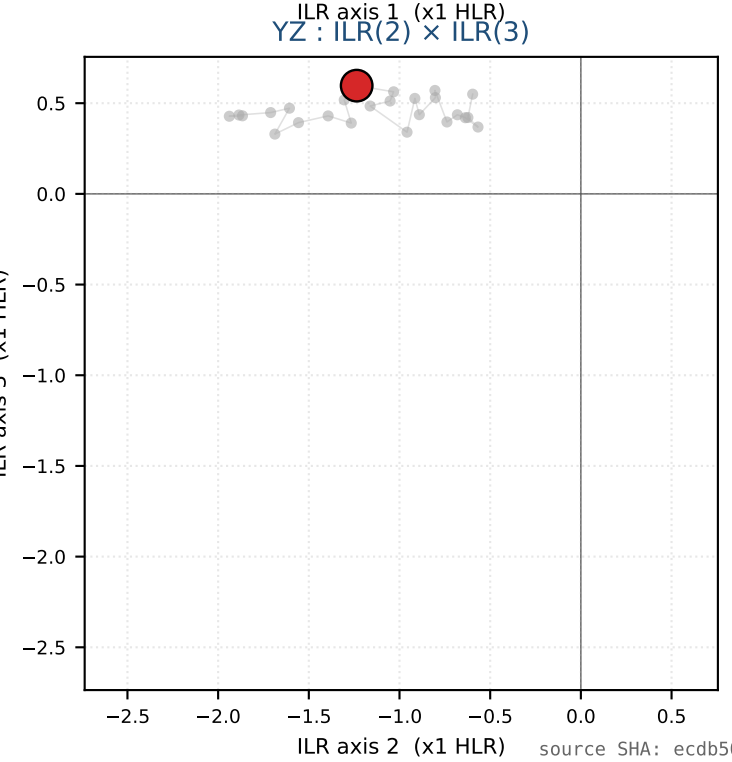
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=14 | 2015 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



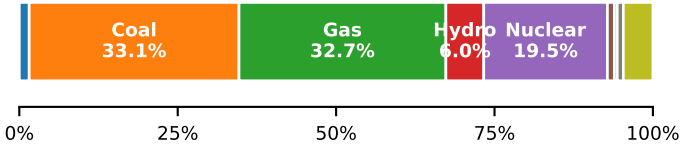




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0156	-1.002
Coal	0.3311	+2.054
Gas	0.3265	+2.040
Hydro	0.0597	+0.342
Nuclear	0.1952	+1.526
Other Fossil	0.0109	-1.361
Other Renewabl	0.0047	-2.205
Solar	0.0096	-1.491
Wind	0.0467	+0.096

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

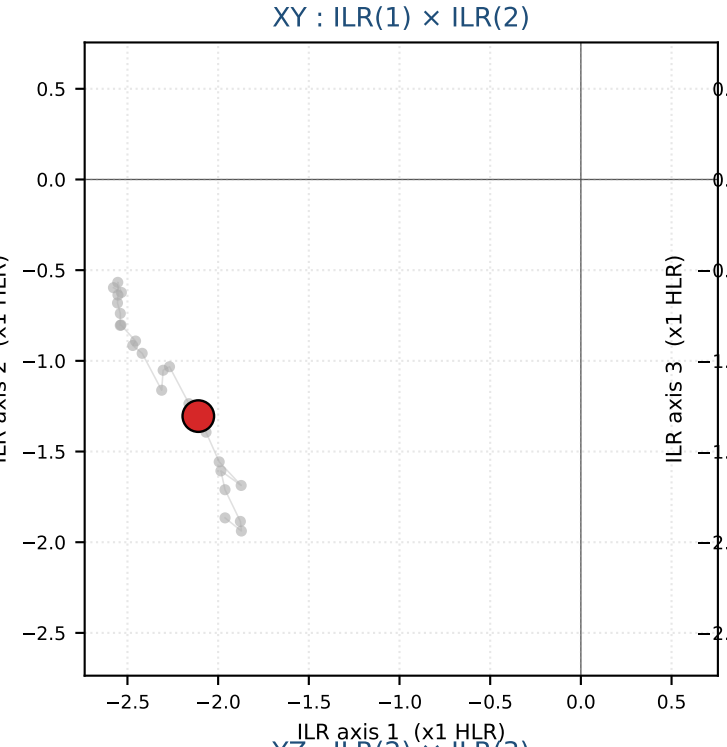
display factor: x1 (raw HLR x 1)

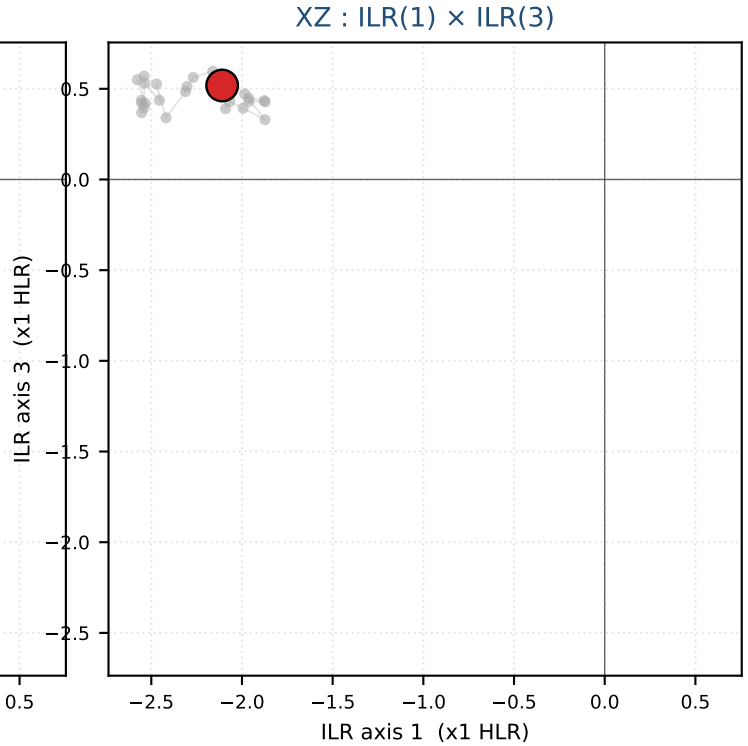
Helmert basis loadings (top 3 carriers per axis)

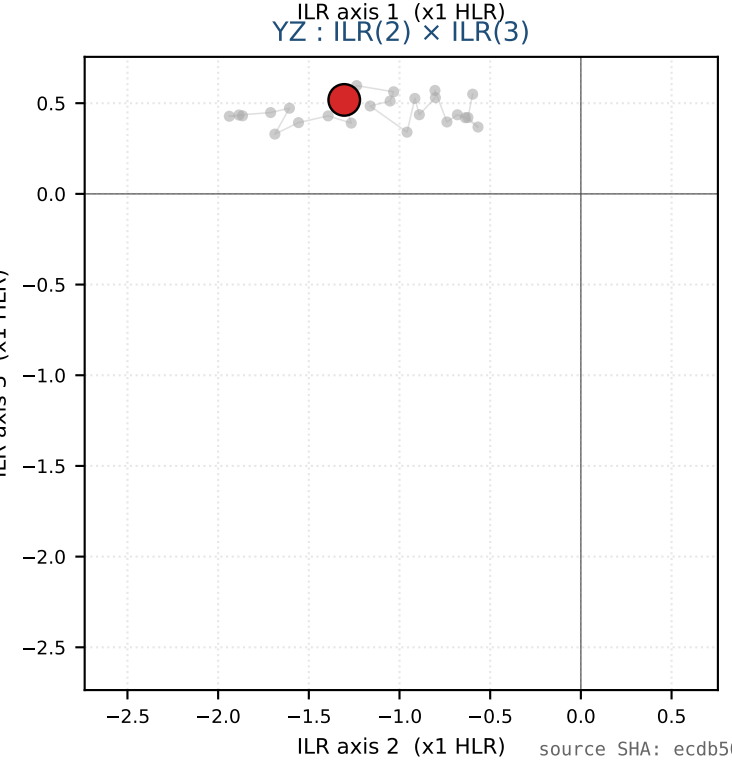
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=15 | 2016 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



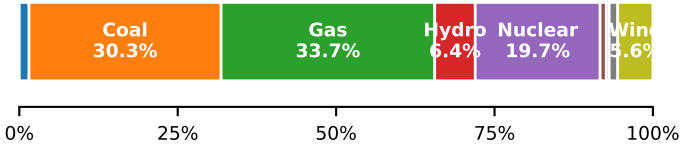




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0154	-1.062
Coal	0.3031	+1.921
Gas	0.3372	+2.028
Hydro	0.0639	+0.364
Nuclear	0.1971	+1.491
Other Fossil	0.0097	-1.524
Other Renewabl	0.0047	-2.247
Solar	0.0134	-1.196
Wind	0.0555	+0.224

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

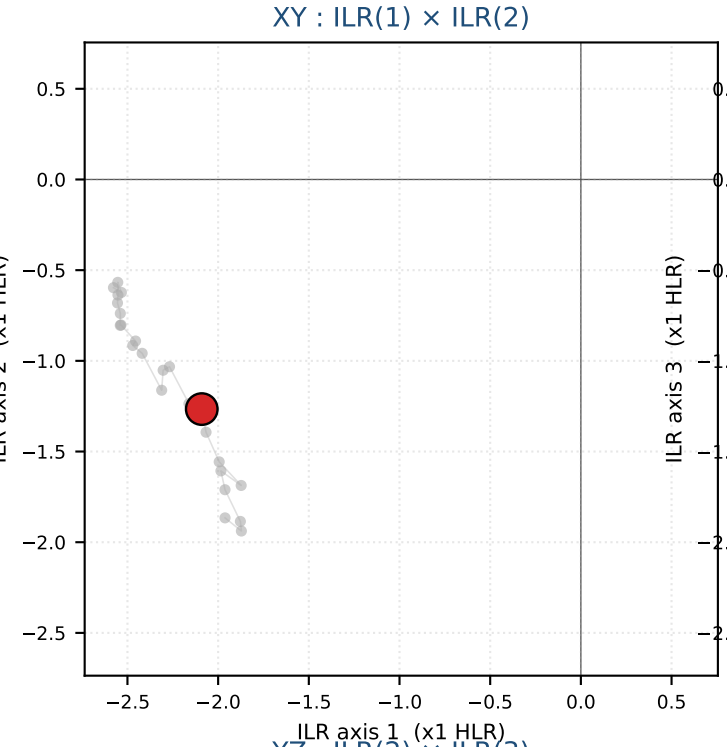
display factor: x1 (raw HLR x 1)

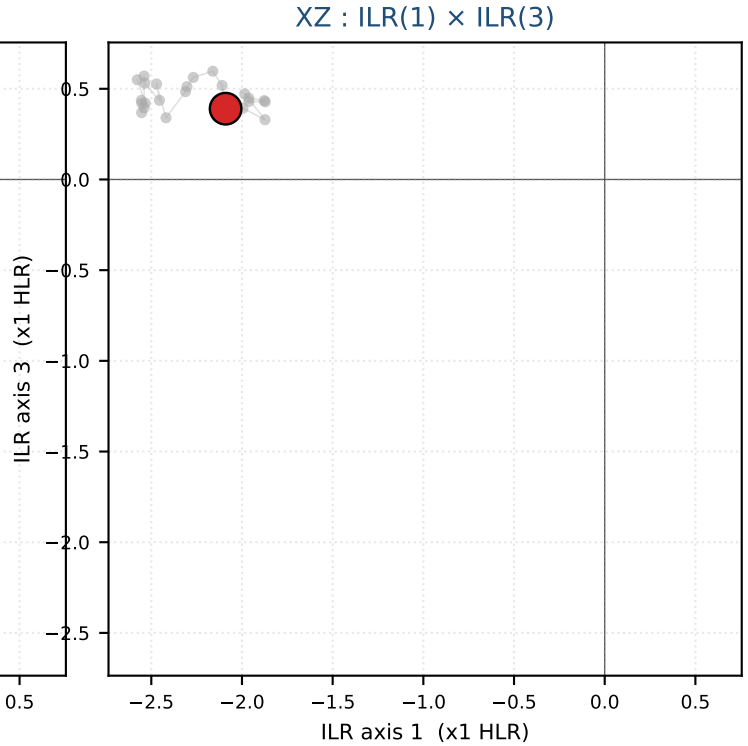
Helmert basis loadings (top 3 carriers per axis)

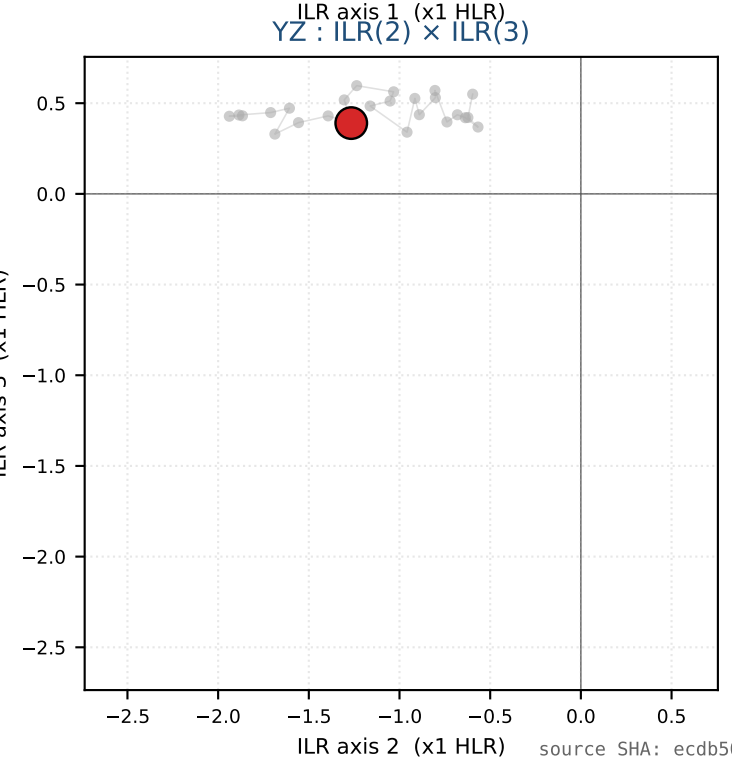
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=16 | 2017 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

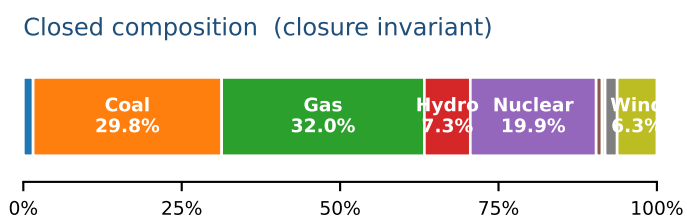






Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0155	-1.110
Coal	0.2976	+1.846
Gas	0.3199	+1.918
Hydro	0.0725	+0.434
Nuclear	0.1986	+1.441
Other Fossil	0.0094	-1.615
Other Renewabl	0.0047	-2.301
Solar	0.0191	-0.902
Wind	0.0628	+0.289



closure: Σ = 1.000000 OK

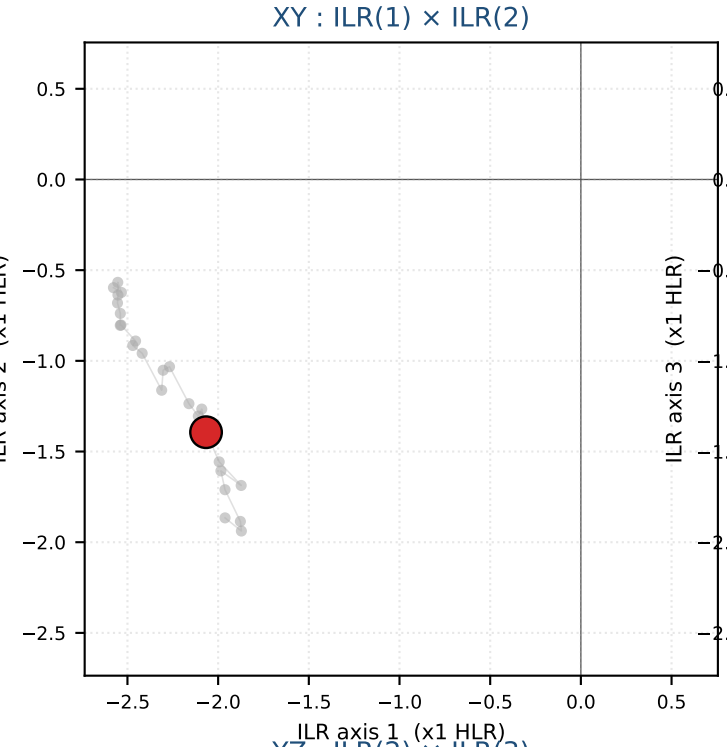
display factor: x1 (raw HLR x 1)

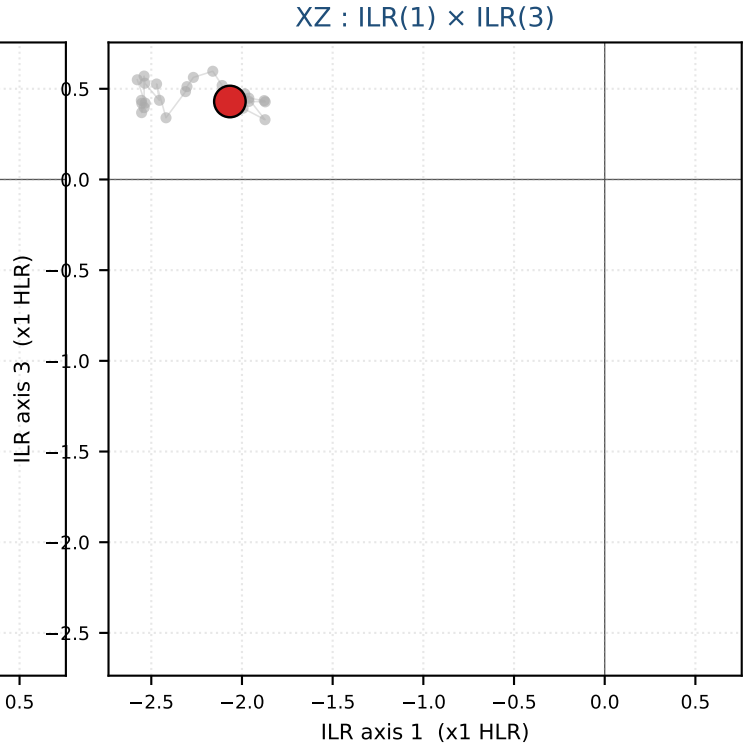
Helmert basis loadings (top 3 carriers per axis)

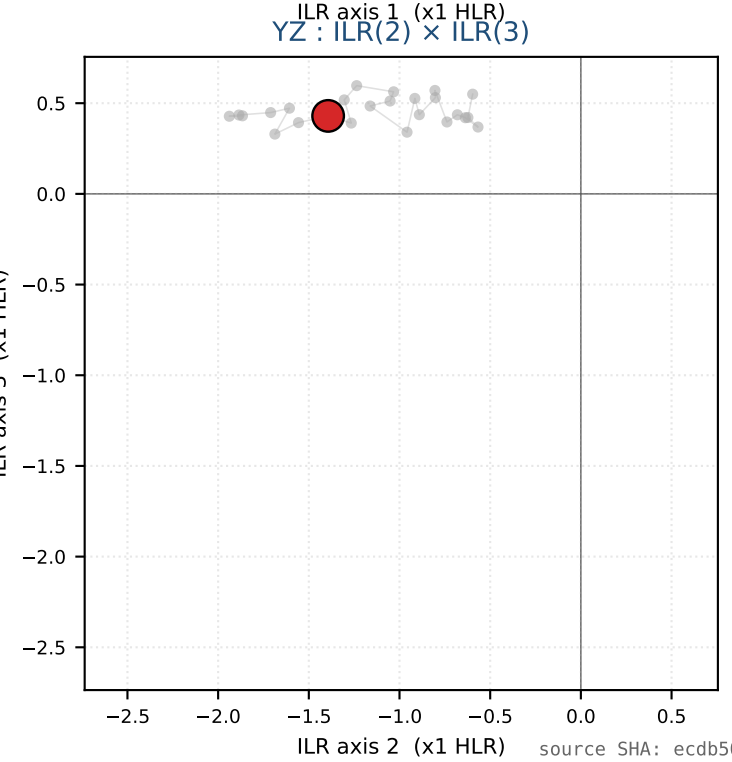
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=17 | 2018 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



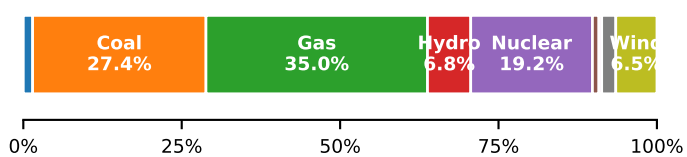




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0147	-1.169
Coal	0.2736	+1.753
Gas	0.3497	+1.999
Hydro	0.0682	+0.365
Nuclear	0.1921	+1.400
Other Fossil	0.0102	-1.540
Other Renewabl	0.0044	-2.366
Solar	0.0222	-0.757
Wind	0.0649	+0.315

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

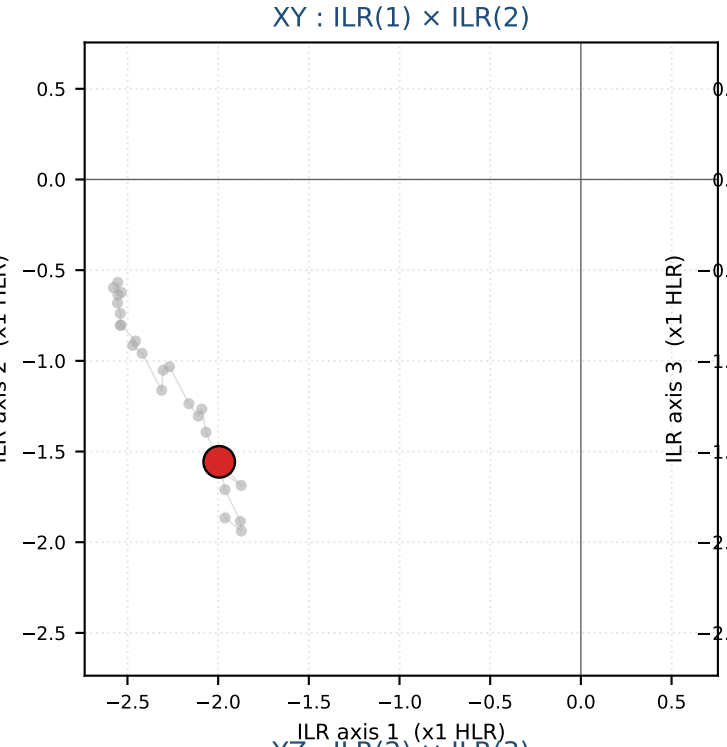
display factor: x1 (raw HLR x 1)

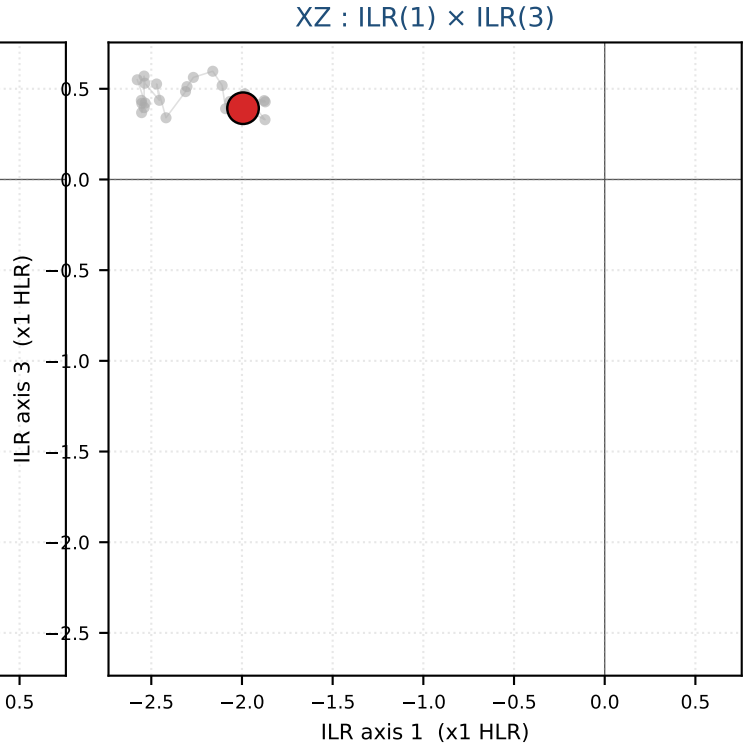
Helmert basis loadings (top 3 carriers per axis)

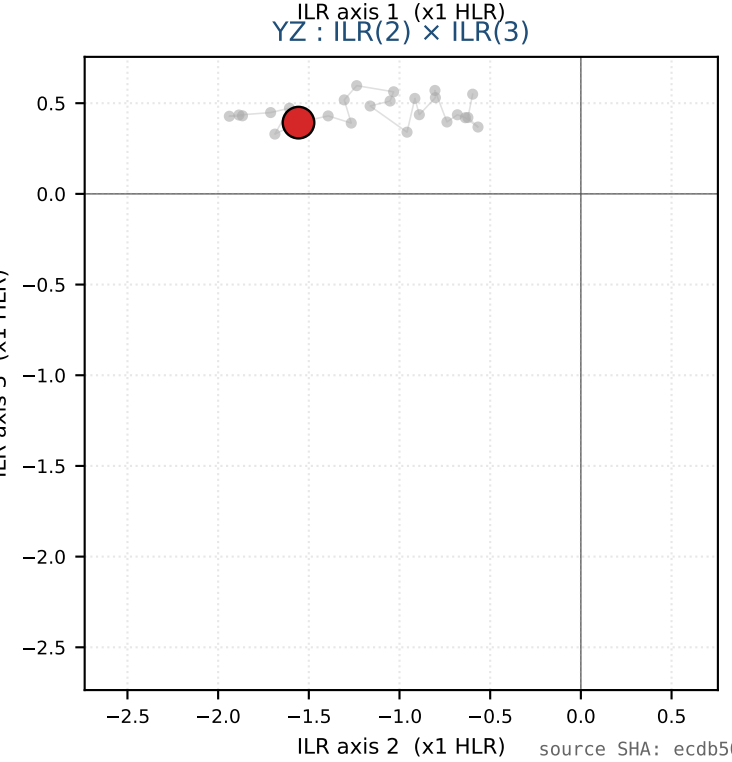
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=18 | 2019 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



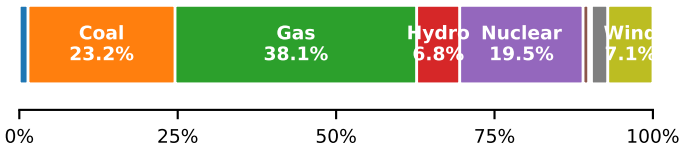




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0138	-1.226
Coal	0.2320	+1.594
Gas	0.3814	+2.091
Hydro	0.0680	+0.366
Nuclear	0.1946	+1.418
Other Fossil	0.0090	-1.654
Other Renewabl	0.0043	-2.394
Solar	0.0257	-0.606
Wind	0.0712	+0.412

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

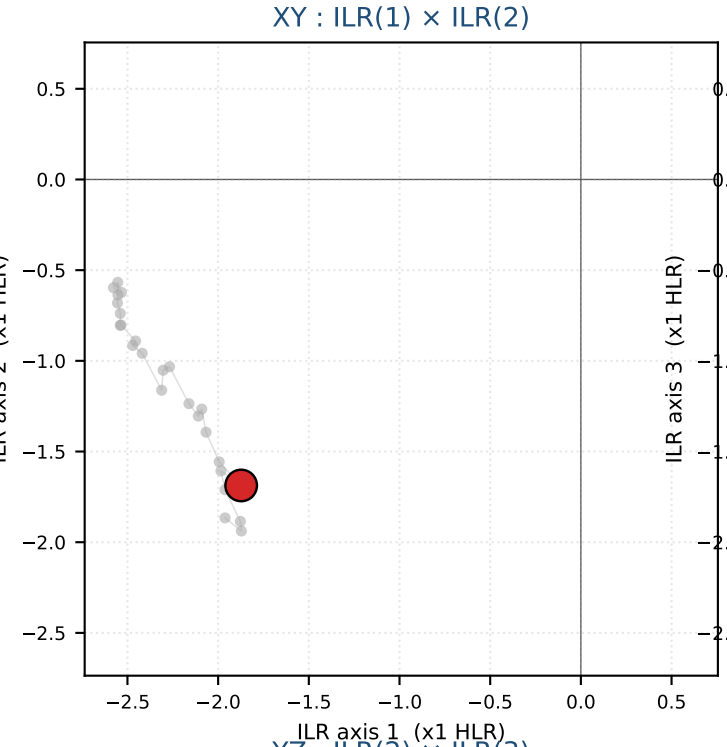
display factor: x1 (raw HLR x 1)

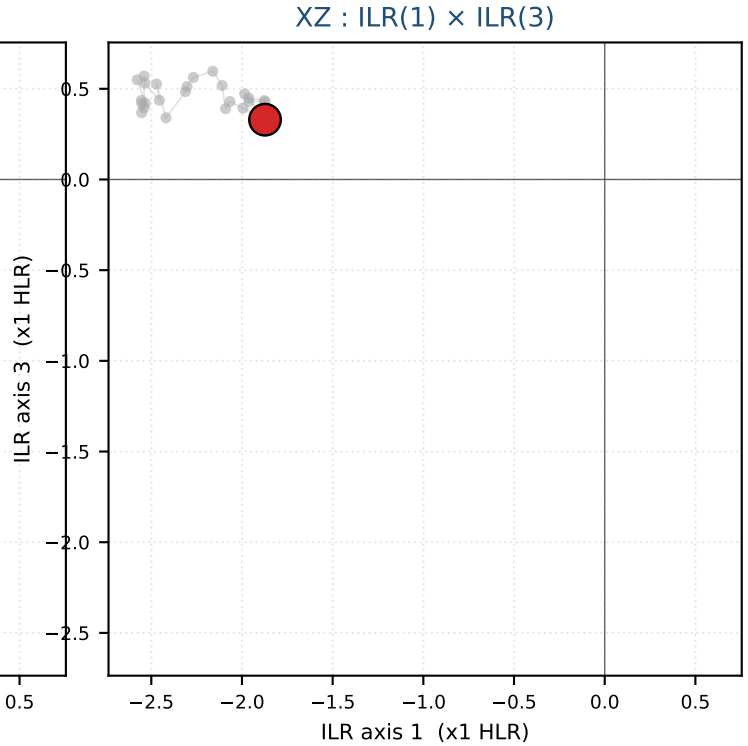
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=19 | 2020 | D=9 T=25 | level 1 first-principles

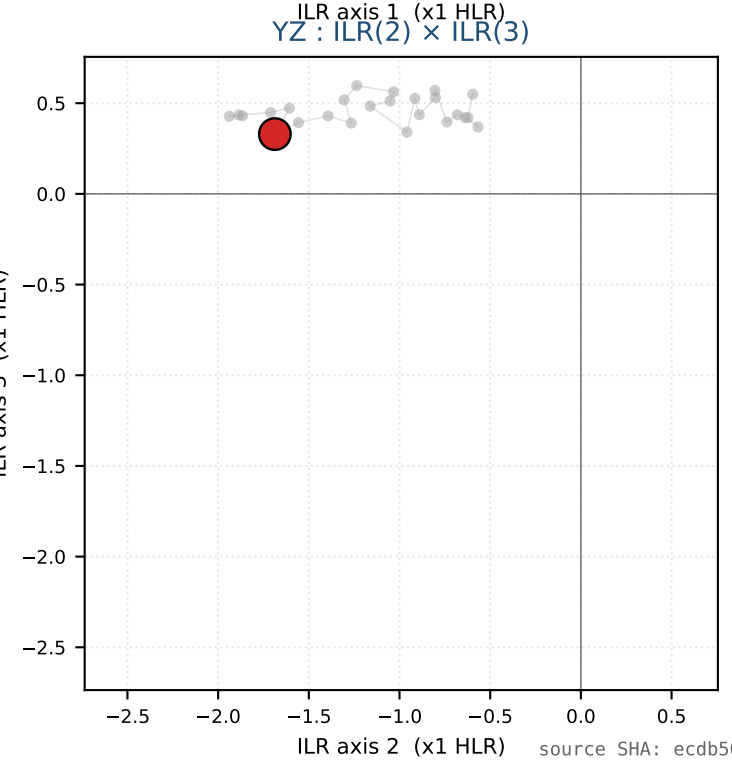
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



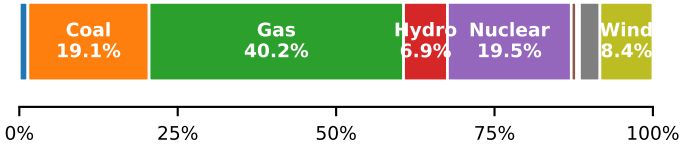


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0135	-1.273
Coal	0.1913	+1.375
Gas	0.4017	+2.117
Hydro	0.0692	+0.359
Nuclear	0.1954	+1.396
Other Fossil	0.0085	-1.739
Other Renewabl	0.0045	-2.381
Solar	0.0323	-0.402
Wind	0.0836	+0.547



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

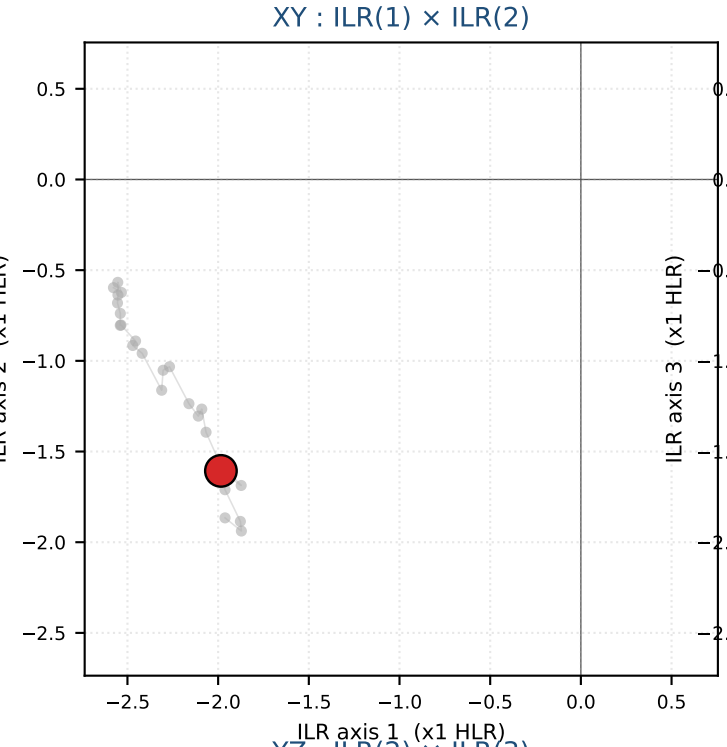
display factor: x1 (raw HLR x 1)

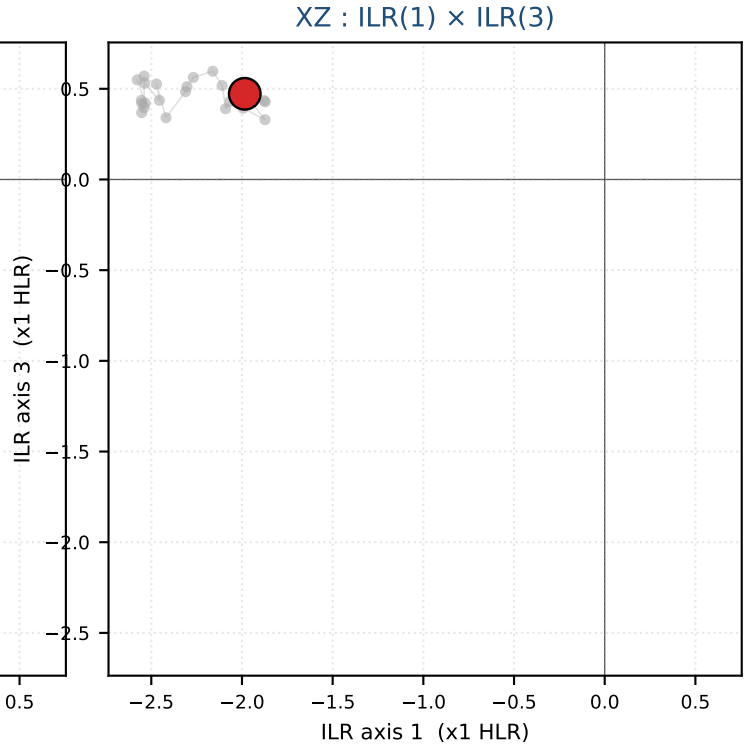
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=20 | 2021 | D=9 T=25 | level 1 first-principles

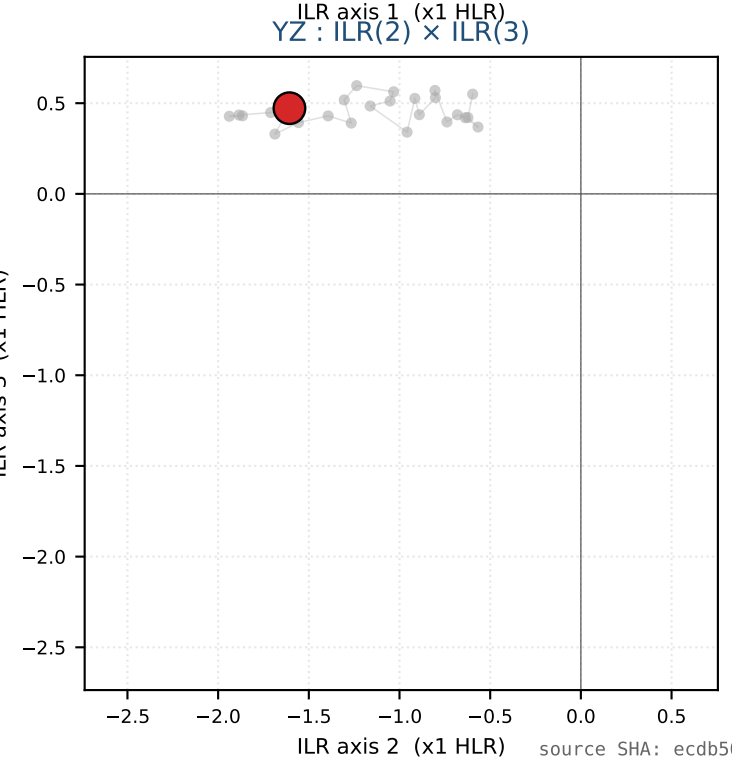
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



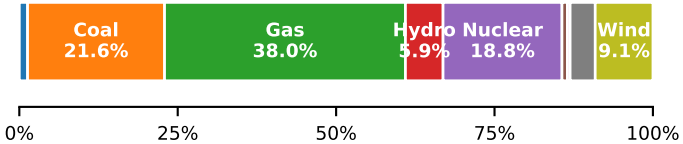


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0131	-1.321
Coal	0.2162	+1.486
Gas	0.3802	+2.050
Hydro	0.0593	+0.193
Nuclear	0.1877	+1.345
Other Fossil	0.0085	-1.753
Other Renewabl	0.0044	-2.409
Solar	0.0396	-0.212
Wind	0.0911	+0.621



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

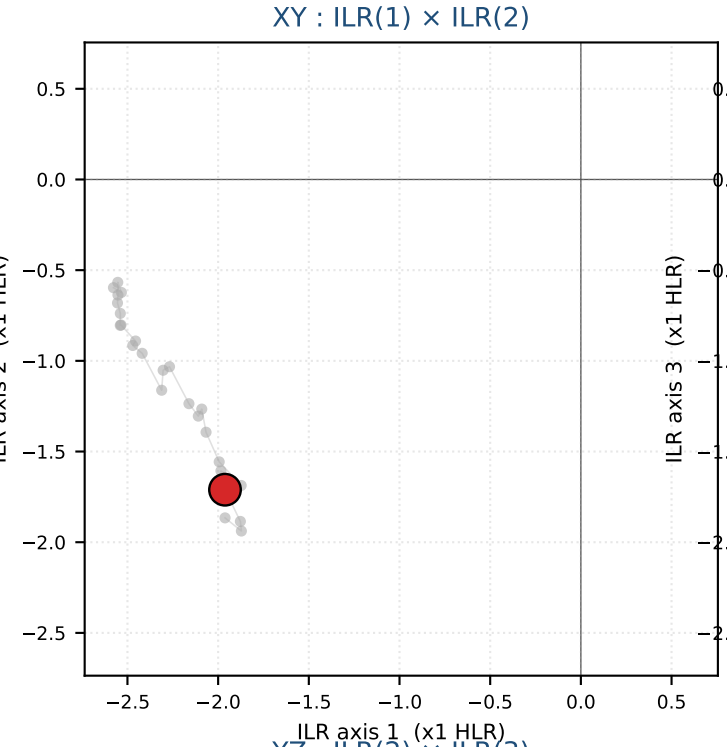
display factor: x1 (raw HLR x 1)

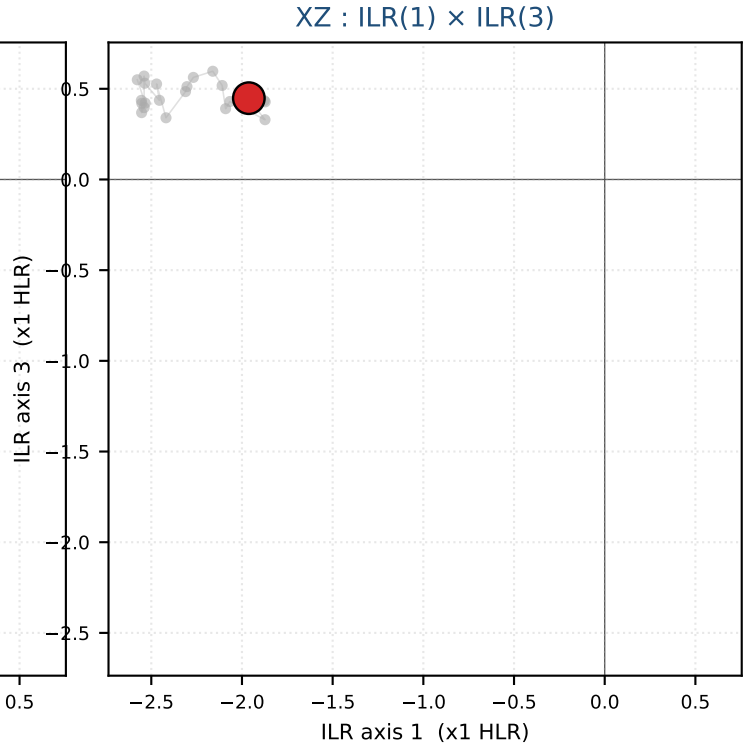
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=21 | 2022 | D=9 T=25 | level 1 first-principles

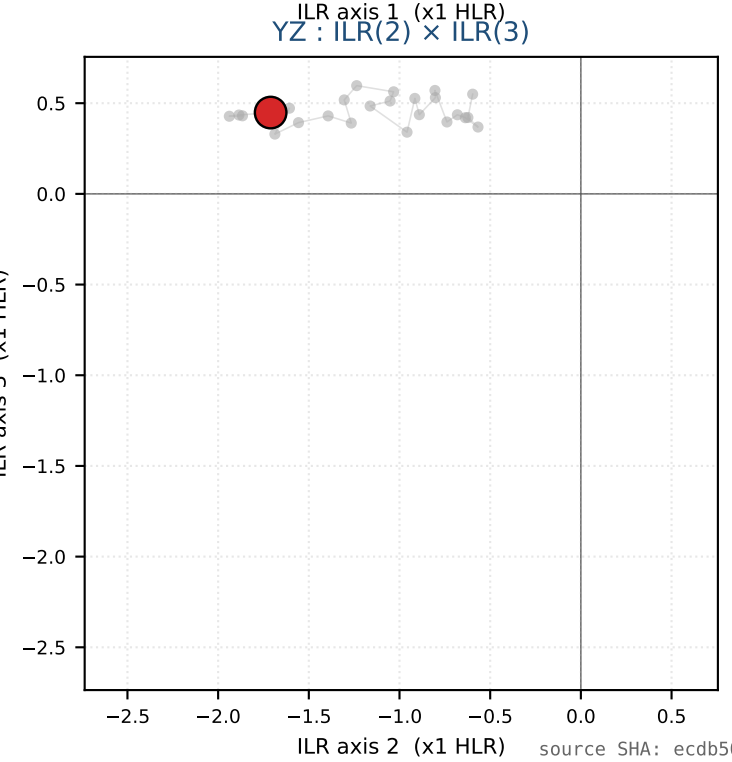
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



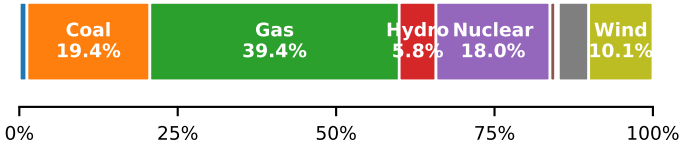


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0121	-1.409
Coal	0.1940	+1.366
Gas	0.3935	+2.073
Hydro	0.0580	+0.159
Nuclear	0.1800	+1.291
Other Fossil	0.0090	-1.703
Other Renewabl	0.0042	-2.459
Solar	0.0478	-0.034
Wind	0.1013	+0.716



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

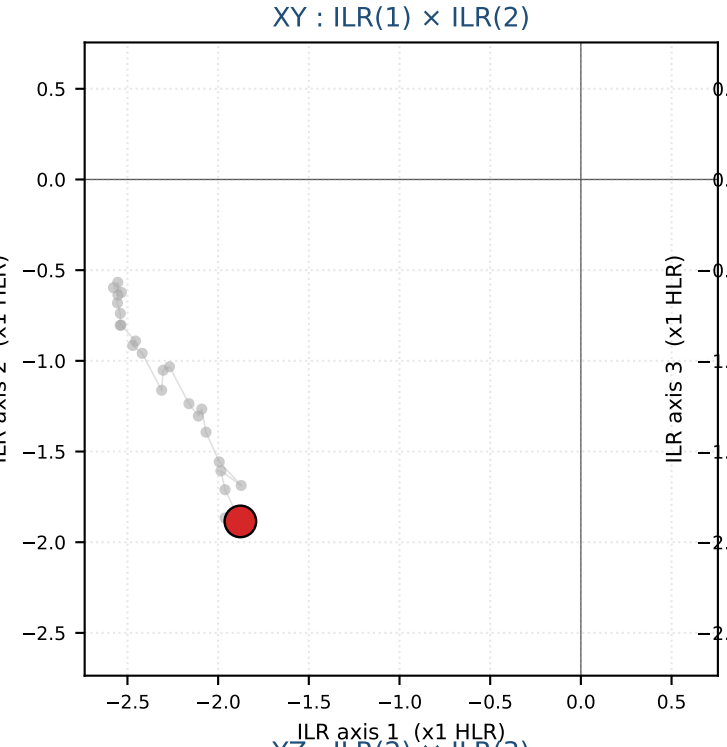
display factor: x1 (raw HLR x 1)

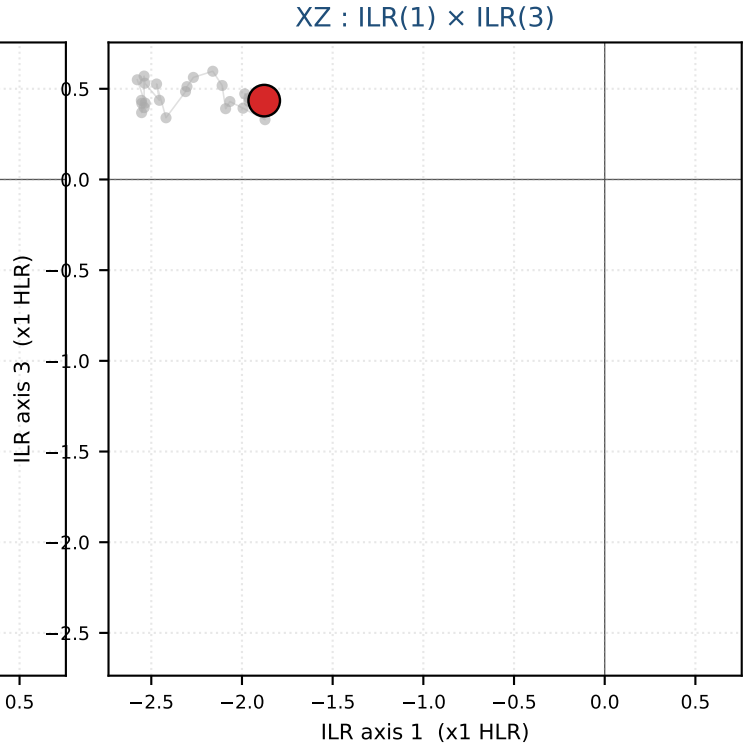
Helmert basis loadings (top 3 carriers per axis)

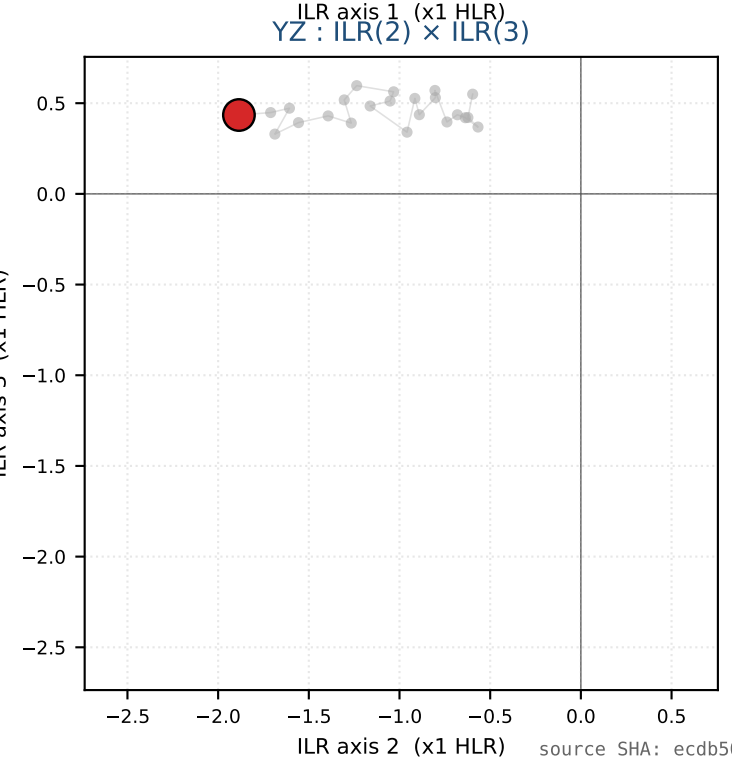
ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=22 | 2023 | D=9 T=25 | level 1 first-principles

ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



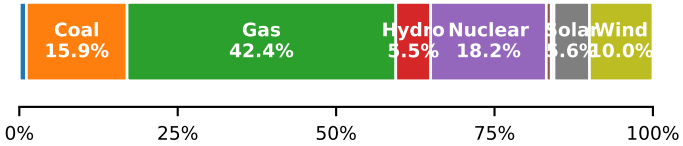




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0112	-1.465
Coal	0.1589	+1.190
Gas	0.4241	+2.172
Hydro	0.0551	+0.130
Nuclear	0.1825	+1.328
Other Fossil	0.0078	-1.829
Other Renewabl	0.0044	-2.401
Solar	0.0560	+0.148
Wind	0.1001	+0.728

Closed composition (closure invariant)



closure: Σ = 1.000000 OK

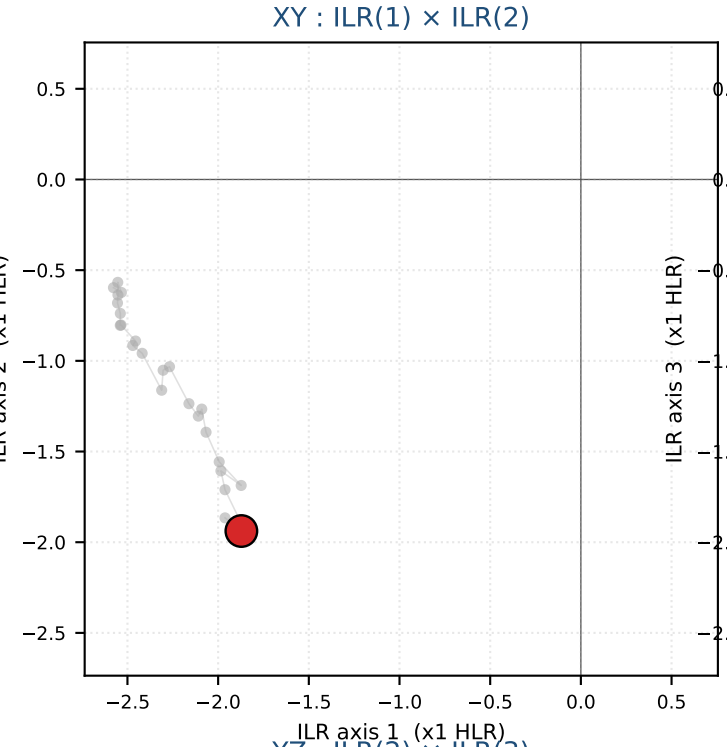
display factor: x1 (raw HLR x 1)

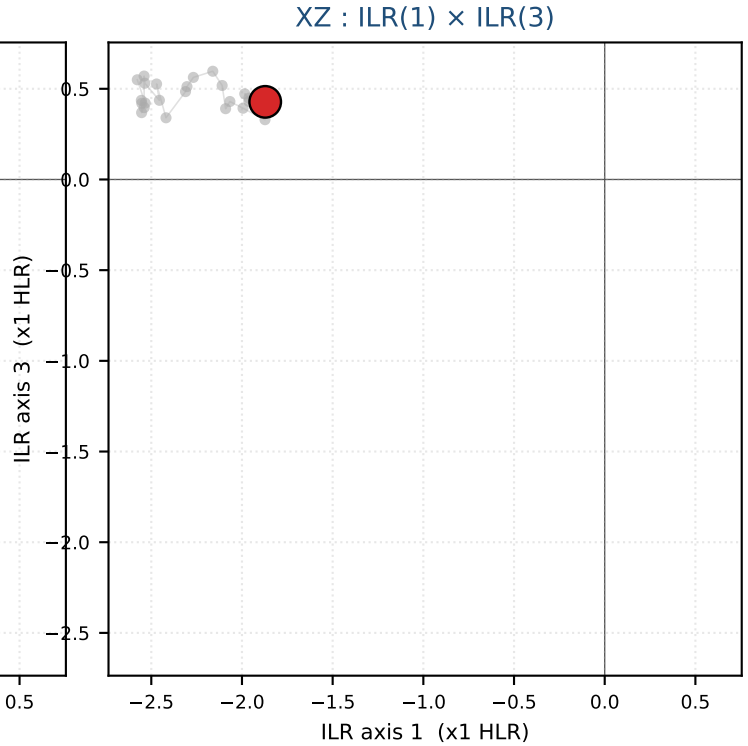
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=23 | 2024 | D=9 T=25 | level 1 first-principles

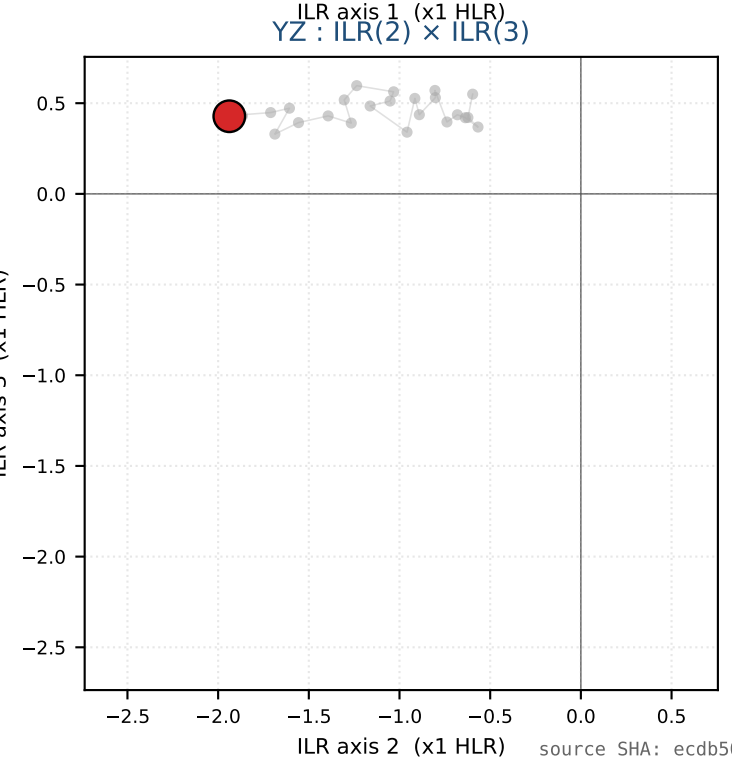
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3

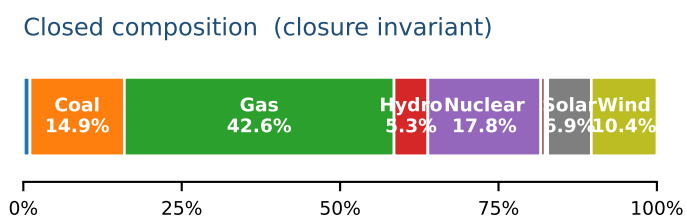




Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0106	-1.505
Coal	0.1489	+1.142
Gas	0.4257	+2.192
Hydro	0.0533	+0.115
Nuclear	0.1783	+1.322
Other Fossil	0.0069	-1.933
Other Renewabl	0.0040	-2.481
Solar	0.0688	+0.370
Wind	0.1036	+0.778





closure: Σ = 1.000000 OK

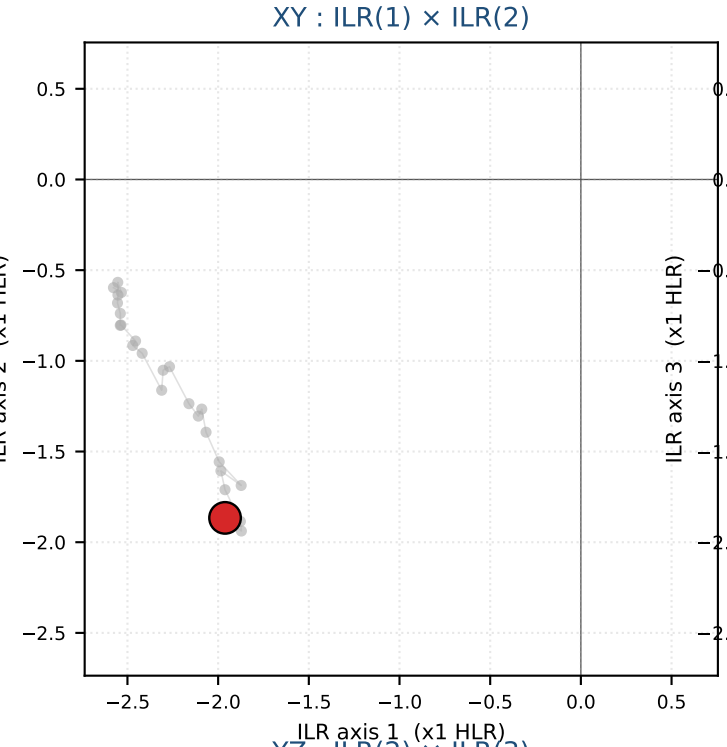
display factor: x1 (raw HLR x 1)

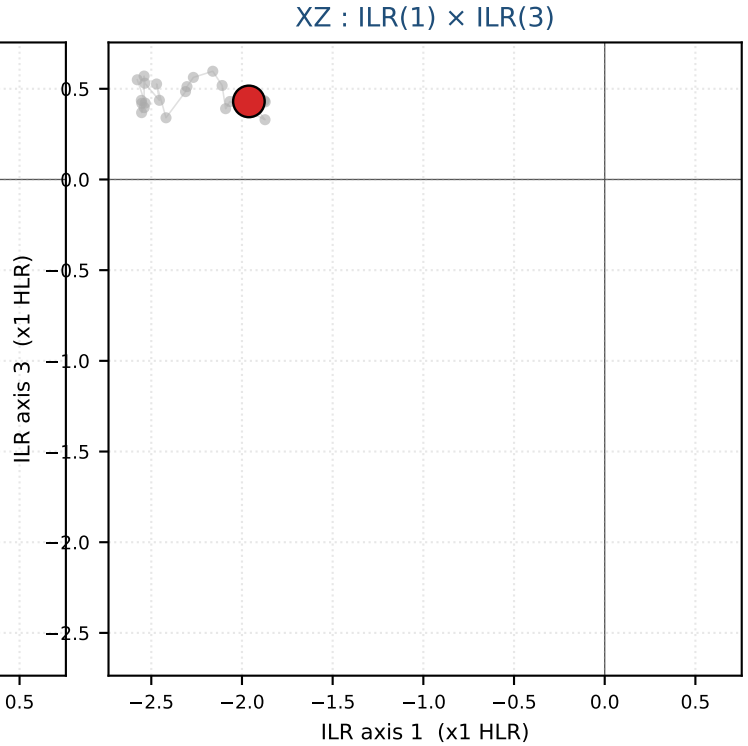
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

STAGE 1 v4 | t=24 | 2025 | D=9 T=25 | level 1 first-principles

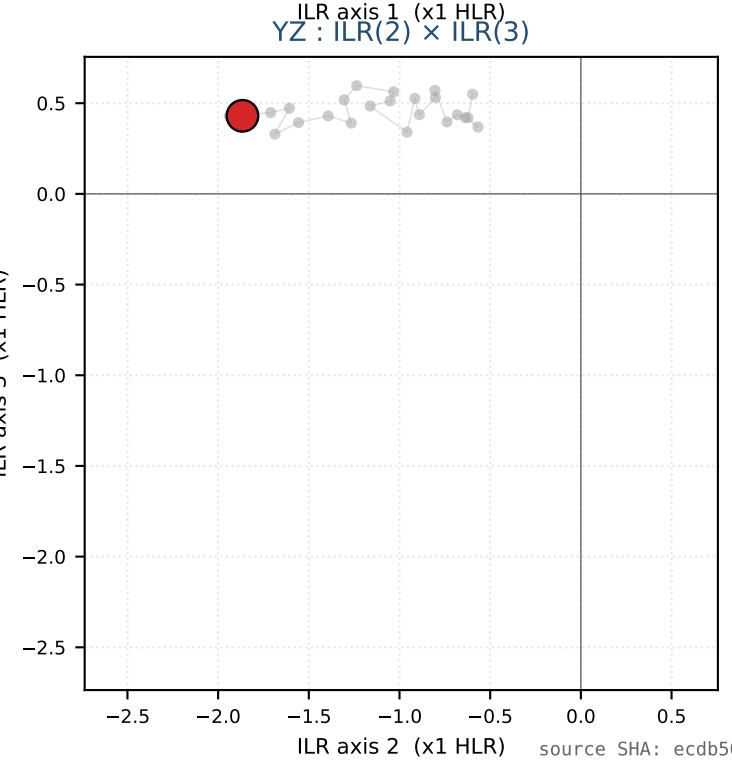
ember_usa | ILR via Helmert basis (orthonormal, HLR units) | display factor x1 | Engine cnt 2.0.3



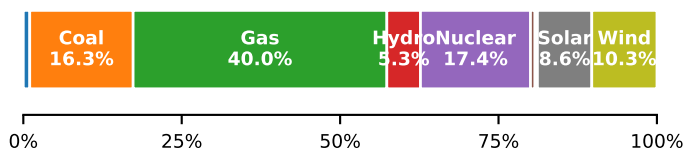


Closed composition (first principles)

carrier	closed	CLR
Bioenergy	0.0102	-1.561
Coal	0.1632	+1.213
Gas	0.4005	+2.111
Hydro	0.0531	+0.091
Nuclear	0.1736	+1.275
Other Fossil	0.0070	-1.929
Other Renewabl	0.0039	-2.521
Solar	0.0858	+0.570
Wind	0.1028	+0.751



Closed composition (closure invariant)



closure: Σ = 1.000000 OK

display factor: x1 (raw HLR x 1)

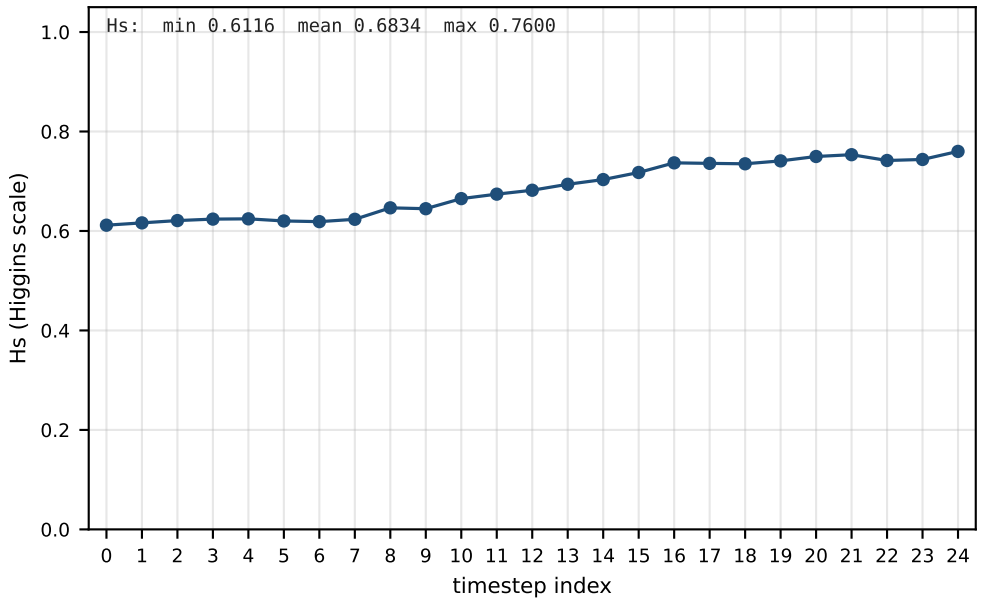
Helmert basis loadings (top 3 carriers per axis)

ILR(1):	Bioenergy+0.707	Coal-0.707	Gas+0.000
ILR(2):	Gas-0.816	Bioenergy+0.408	Coal+0.408
ILR(3):	Hydro-0.866	Bioenergy+0.289	Coal+0.289

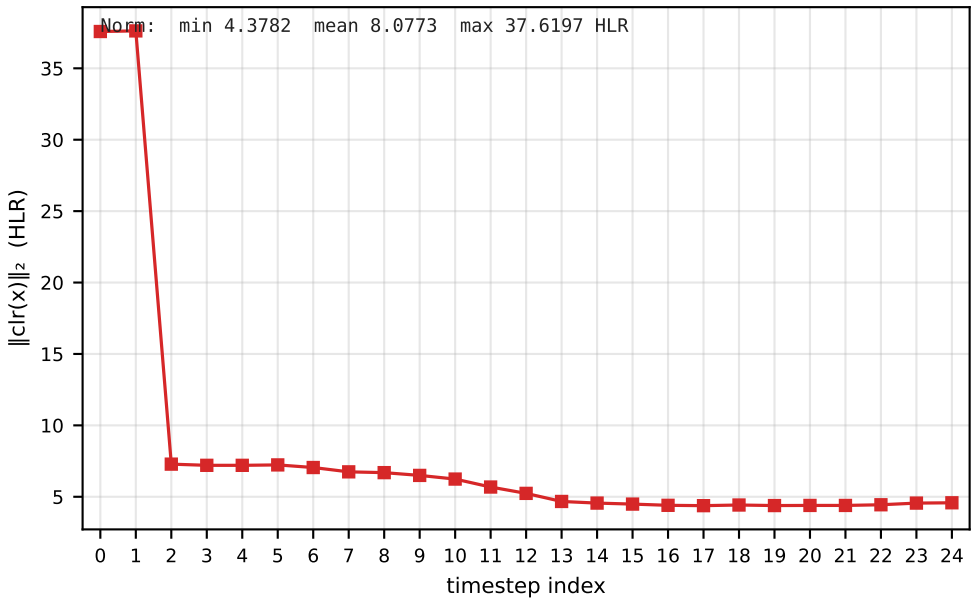
STAGE 1 SUMMARY | ember_usa | D=9 T=25 | Order 1 (first-principles closure)

Engine cnt 2.0.3 | Schema 2.0.0 | Generated 2026-05-05T20:04:43

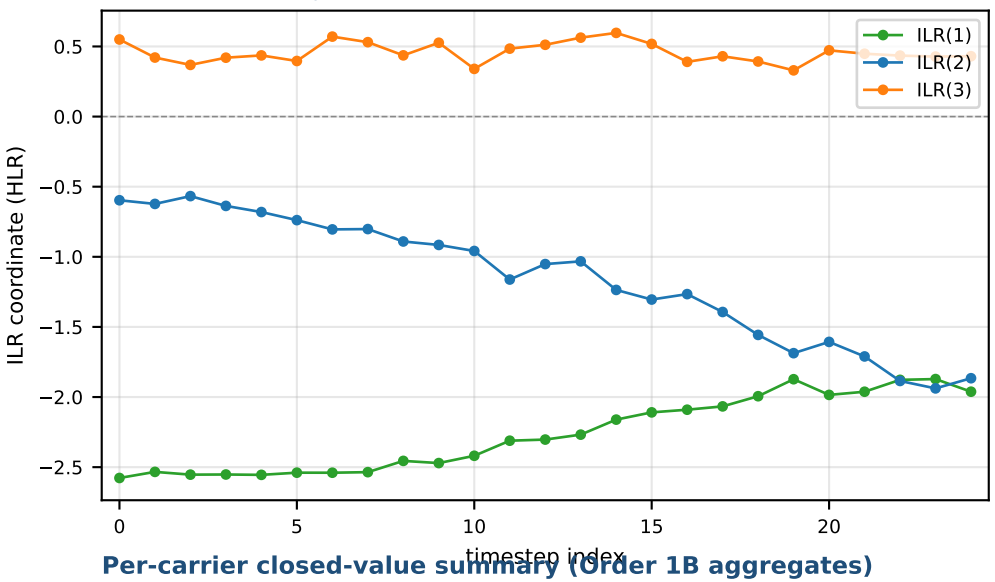
Hs trajectory — single-composition entropy / ln(D)



Aitchison norm — distance from centroid (HLR)



ILR axes 1–3 trajectories (Helmert basis)



Order-1 invariants (whole trajectory)

```
closure invariant     :  max |Σ closed − 1| = 2.22e-16
closure status        :  OK ✓

Hs range              :  [0.6116, 0.7600]
Aitchison norm range  :  [4.3782, 37.6197] HLR
ILR(1) range          :  [-2.5772, -1.8718]
ILR(2) range          :  [-1.9383, -0.5672]
ILR(3) range          :  [+0.3296, +0.5968]

Source file           :  ember_USA_United_States_generation_TWh.csv
Source SHA-256        :  ecdb50b25cb5807b38d4767a0344eddf...
Closed-data SHA-256   :  81879347e313f033e024b2bc26041b8b...
Content SHA-256       :  fae4ef547caa28e8775143a72b9013b7...
Ordering              :  temporal=True, method=by-time
Zero replacement δ    :  1e-15
```

Per-carrier closed-value summary (Order 1B aggregates)

#	carrier	min	mean	max	std
0	Bioenergy	0.0102	0.0136	0.0156	0.0014
1	Coal	0.1489	0.3586	0.5109	0.1276
2	Gas	0.1680	0.2895	0.4257	0.0864
3	Hydro	0.0531	0.0637	0.0765	0.0060
4	Nuclear	0.1736	0.1933	0.2063	0.0076
5	Other Fossil	0.0069	0.0149	0.0364	0.0092
6	Other Renewabl	0.0039	0.0043	0.0047	0.0002
7	Solar	0.0000	0.0173	0.0858	0.0241
8	Wind	0.0018	0.0448	0.1036	0.0359